

24m0847 CS790 Assignment 1 Learnings

Issues

Text editor on Free BSD?

- FreeBSD also comes with more powerful text editors such as vi as part of the base system, while other editors, like **emacs** and **vim**, are part of the FreeBSD Ports Collection

Configure internet in freebsd for installing pkgs

- <https://docs.freebsd.org/en/books/handbook/network/>

enable internet in virtualbox

- open the settings of your virtual machine, navigate to the "Network" section, ensure the network adapter is enabled, and select "Attached to: Bridged Adapter" to directly connect your virtual machine to your host computer's network, granting it internet access

Install Sudo in freebsd using root

```
su
pkg install sudo
```

User not in Sudoers

```
$ su - # visudo
```

Append the following line and exit from a text editor:

```
alice ALL=(ALL) ALLin freebsd
```

Install nano

```
sudo pkg install nano
```

Enable pf

```
nano etc/rc.conf
```

- Add following lines to file

```
pf_enable="YES"           # enable pf (load module if required)
pflog_enable="YES"        # start pflogd(8)
```

Write permission issue

```
sudo nano etc/rc.conf
```

Enable PF

```
sudo kldload pf
sudo pfctl -e
```

```
vansil@firewall:~ $ sudo kldload pf
sudo: kldload: command not found
vansil@firewall:~ $ sudo kldload pf
vansil@firewall:~ $ sudo pfctl -e
pf enabled
vansil@firewall:~ $ █
```

Install `tree`:

```
sudo pkg install tree
```

Create `/etc/pf.conf` file

```
touch pf.conf # permission denied
sudo touch pf.conf
```

```
vansil@firewall:/etc $ ls | grep pf
authpf
pf.os
vansil@firewall:/etc $ touch pf.conf
touch: pf.conf: Permission denied
vansil@firewall:/etc $ sudo touch pf.conf
vansil@firewall:/etc $ █
```

Start pf

```
sudo /etc/rc.d/pf start
```

```
vansil@firewall:/etc $ sudo /etc/rc.d/pf start  
Enabling pf.
```

Stop pf

```
sudo /etc/rc.d/pf stop
```

```
vansil@firewall:/etc $ sudo /etc/rc.d/pf stop  
Disabling pf.
```

Shutdown freeBSD

```
sudo shutdown -p now
```

Task 1:

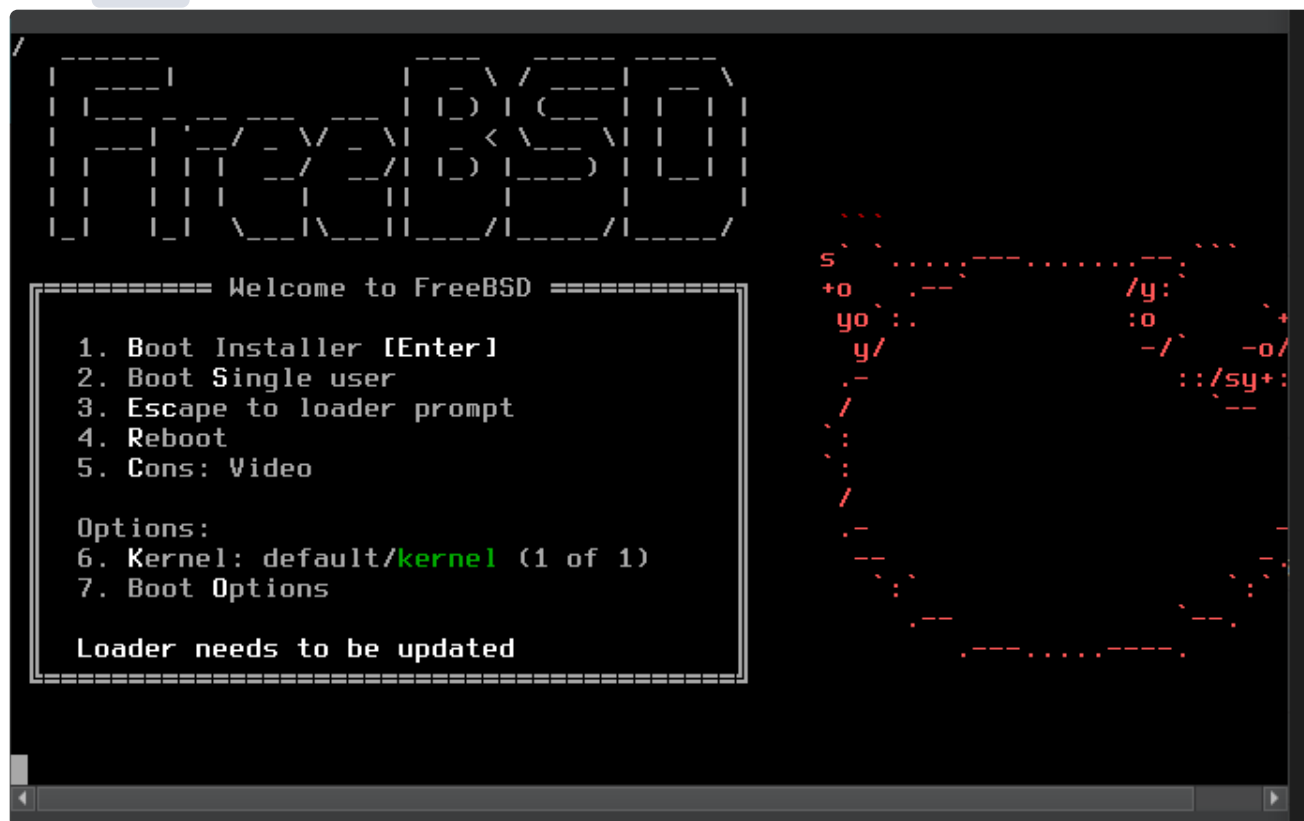
Requirements:

Create 3 VMs.

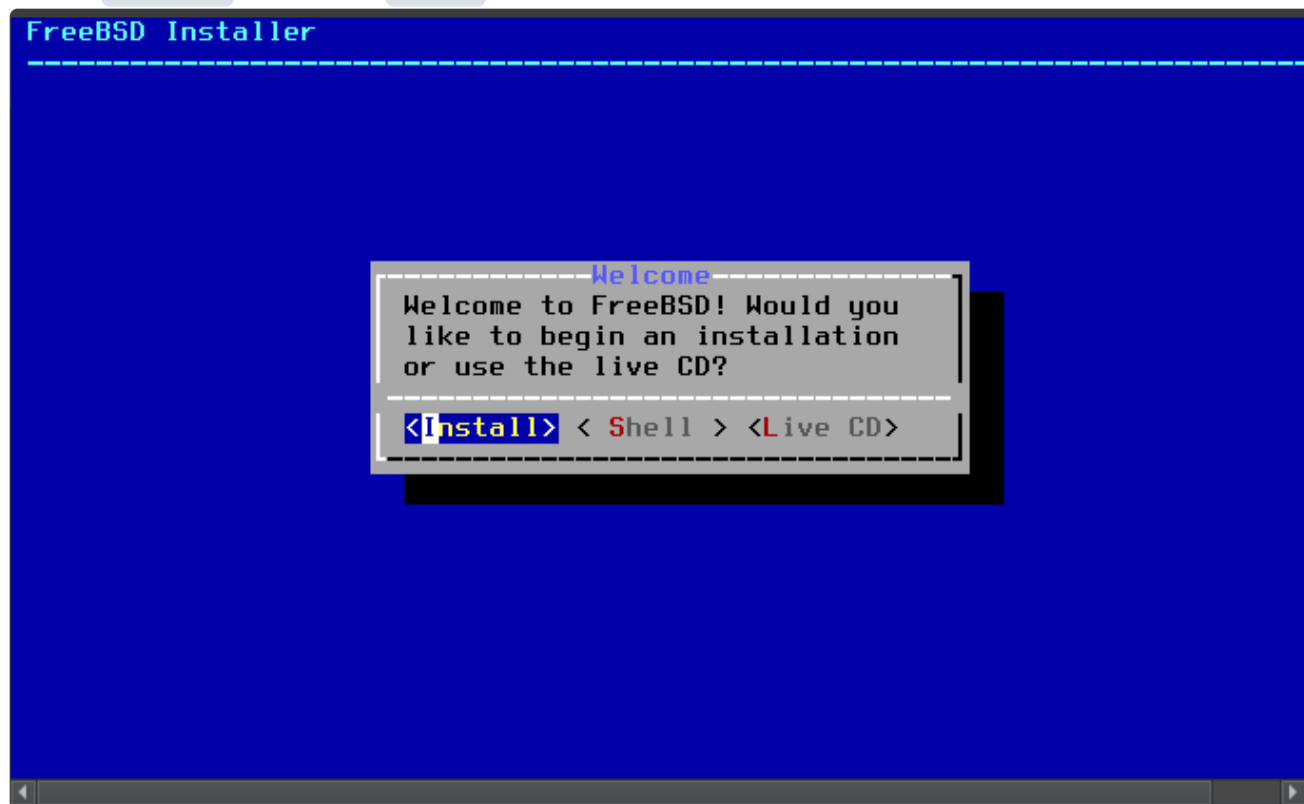
1. FirewallBSD (FreeBSD) (VM2)
2. VM1
3. VM3

Installation:

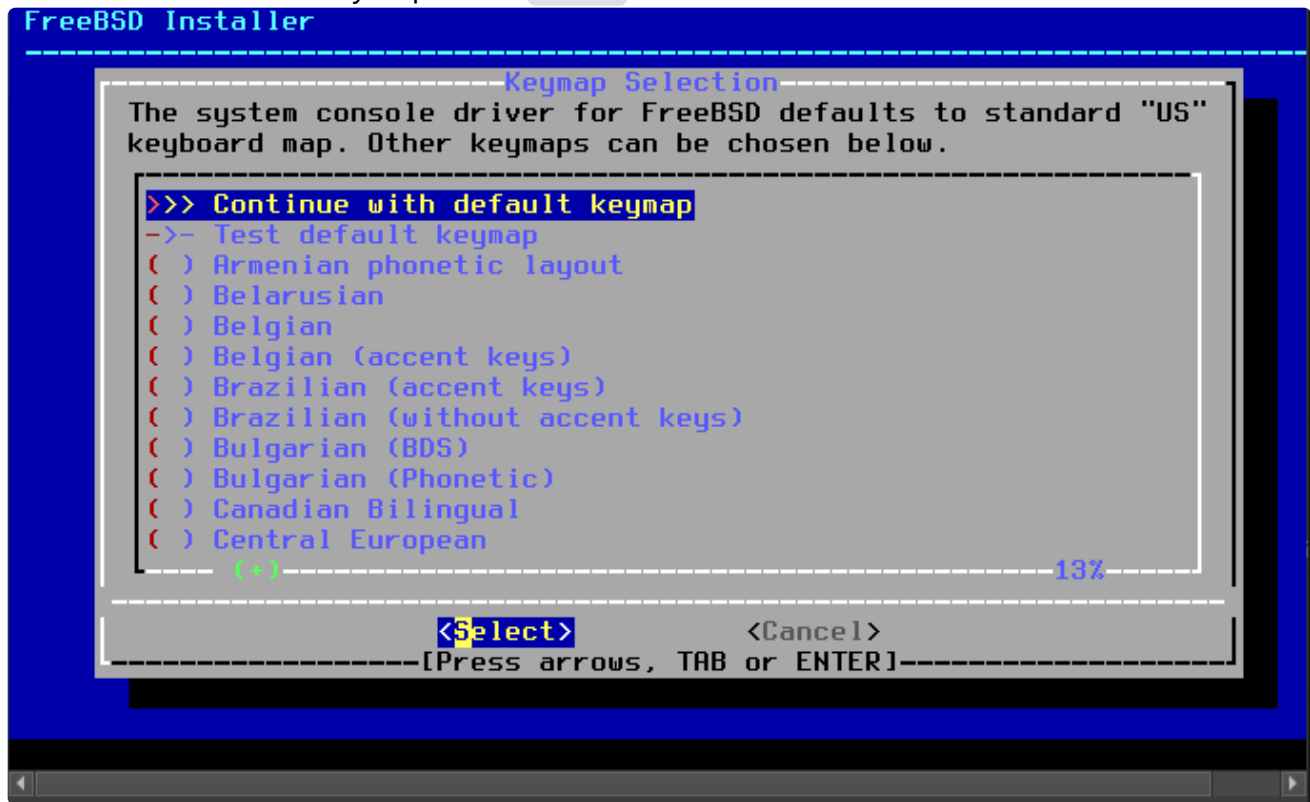
- Press **Enter**



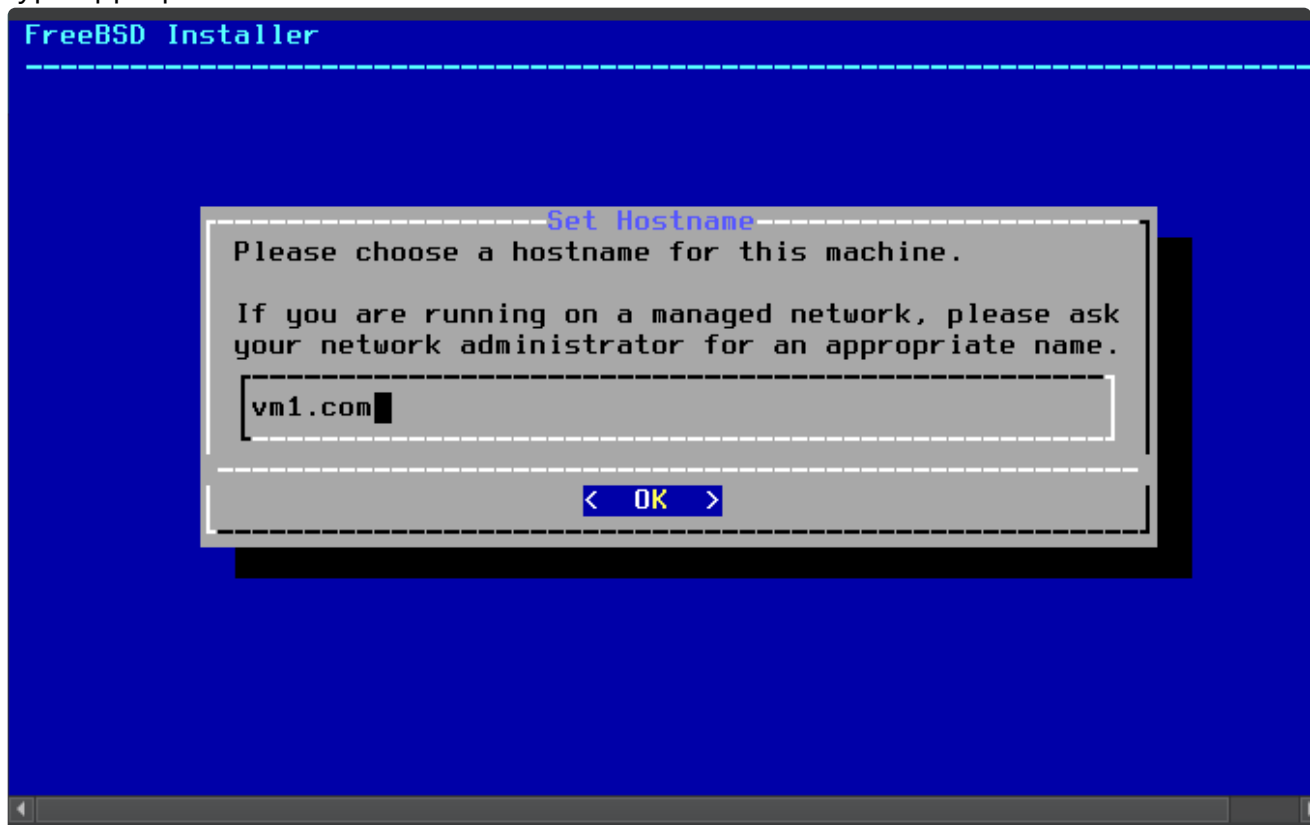
- Select **Install** and Press **Enter**



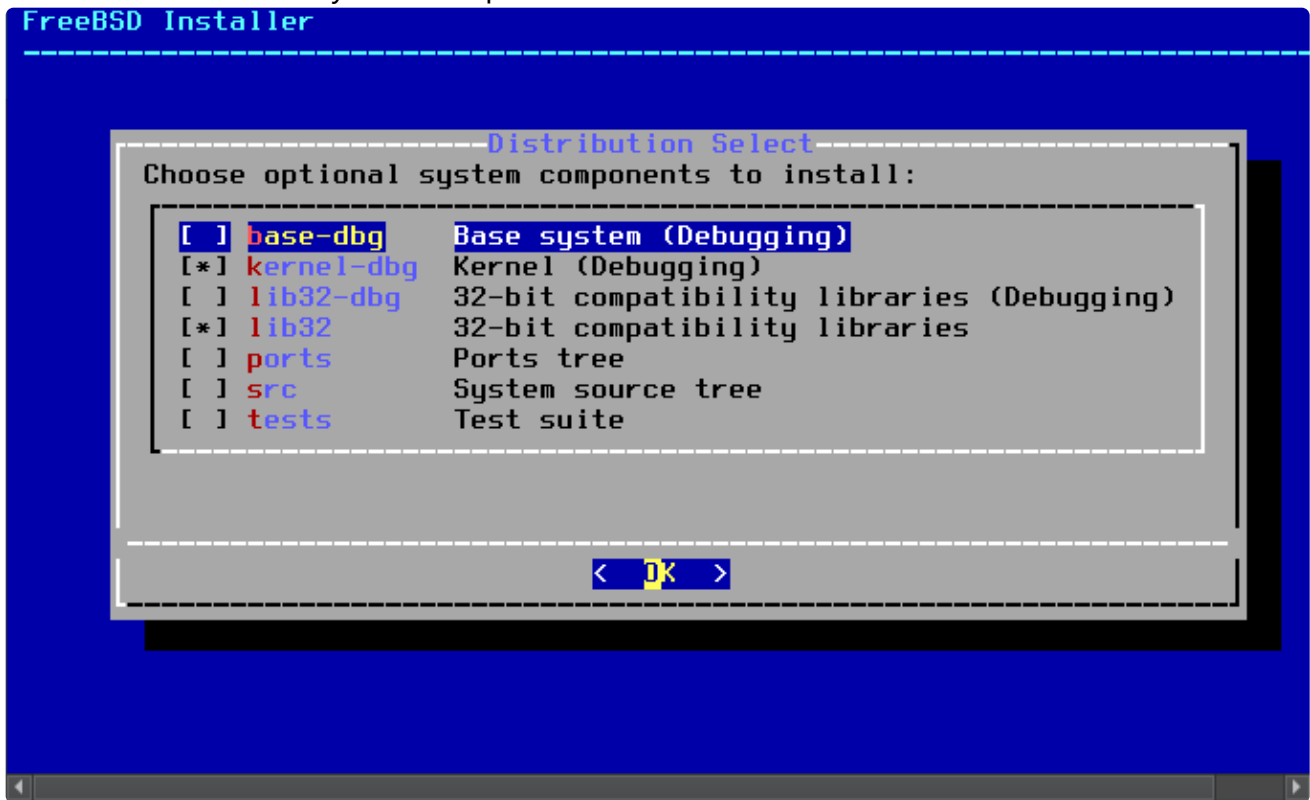
- Continue with default keymap. Press **Enter**



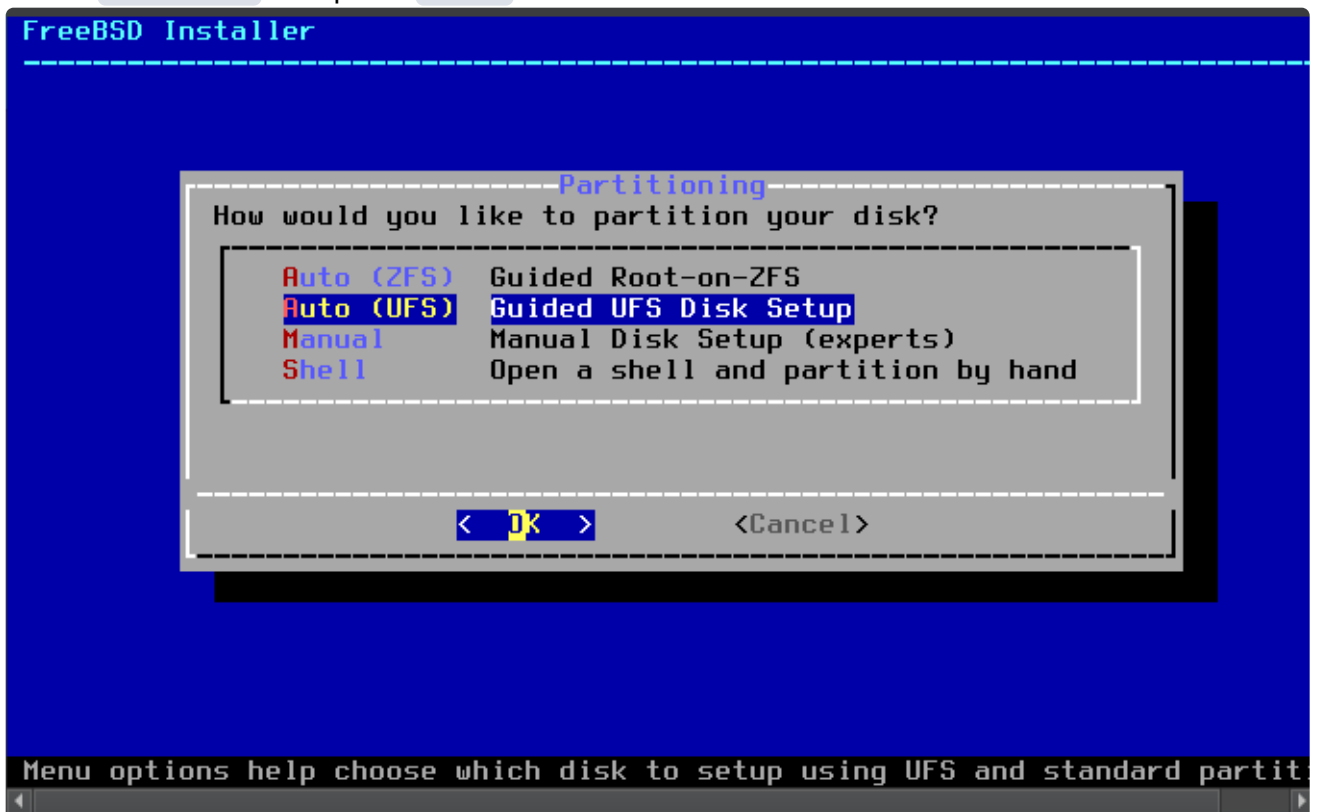
- type appropriate host name.



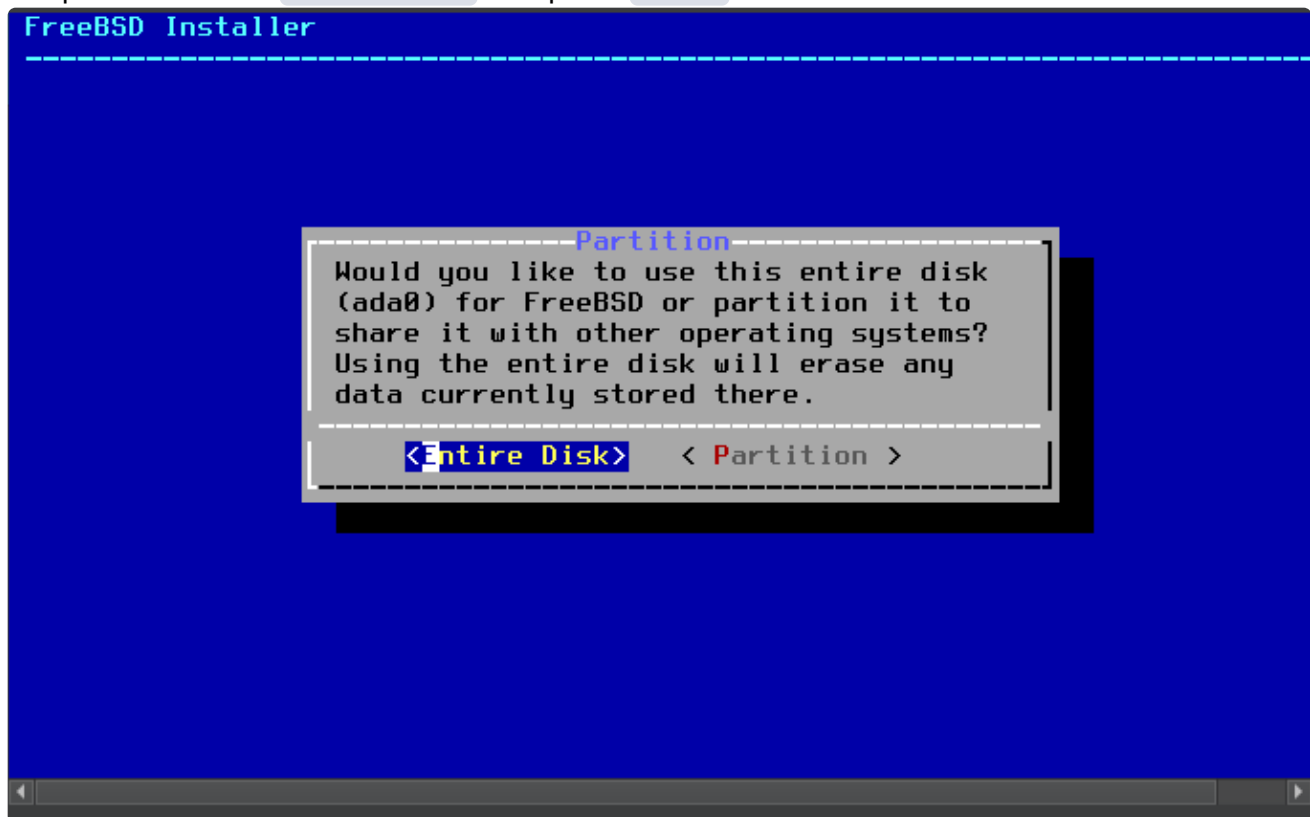
- Continue with default system component selection.



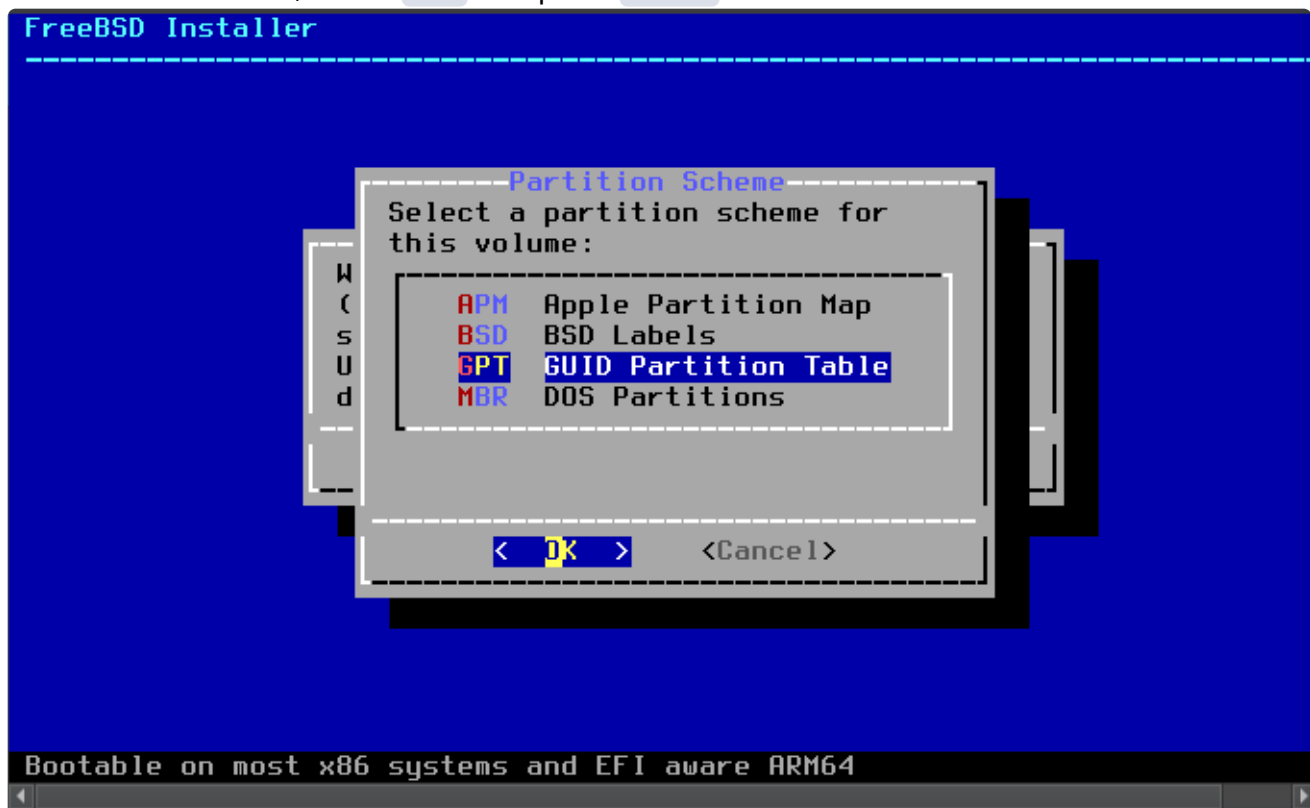
- Select Auto(UFS) and press Enter



- For partition select **Entire Disk** and press **Enter**



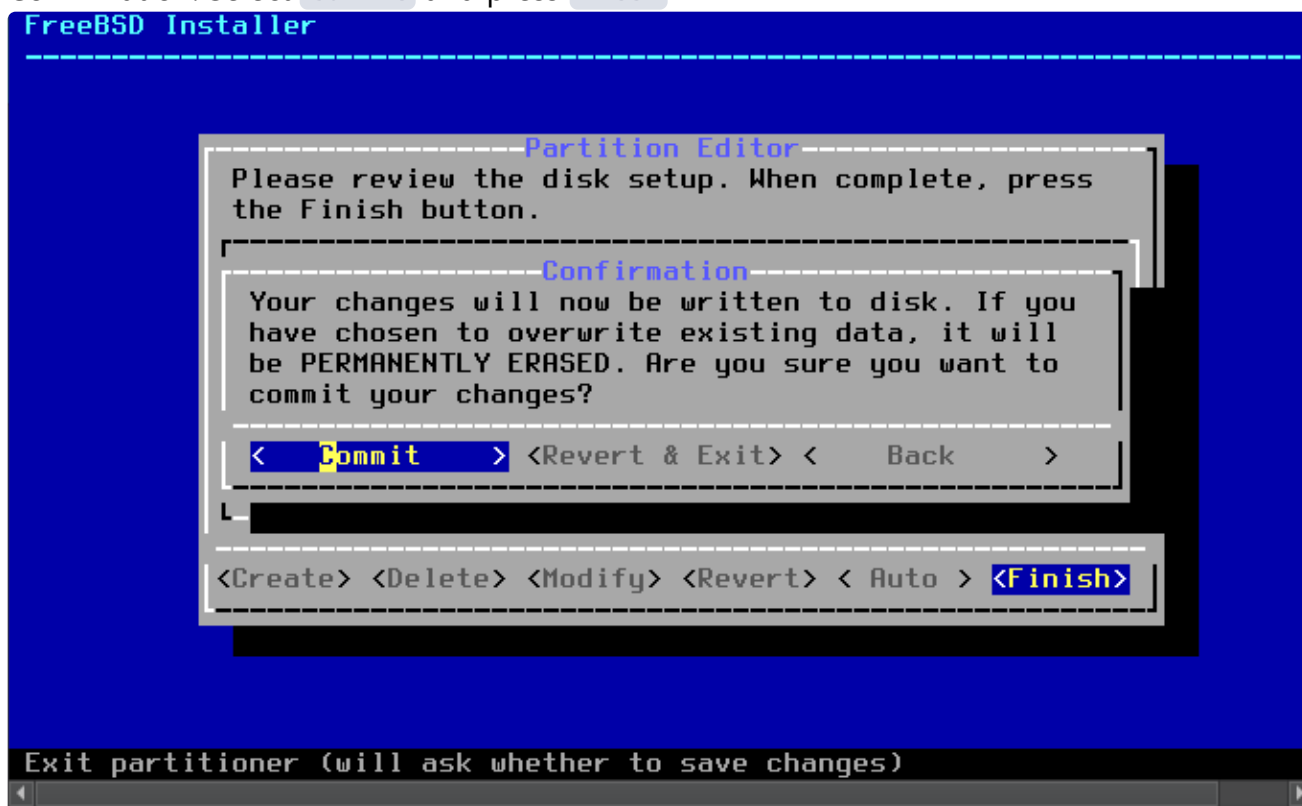
- For Partition scheme, select **GPT** and press **Enter**



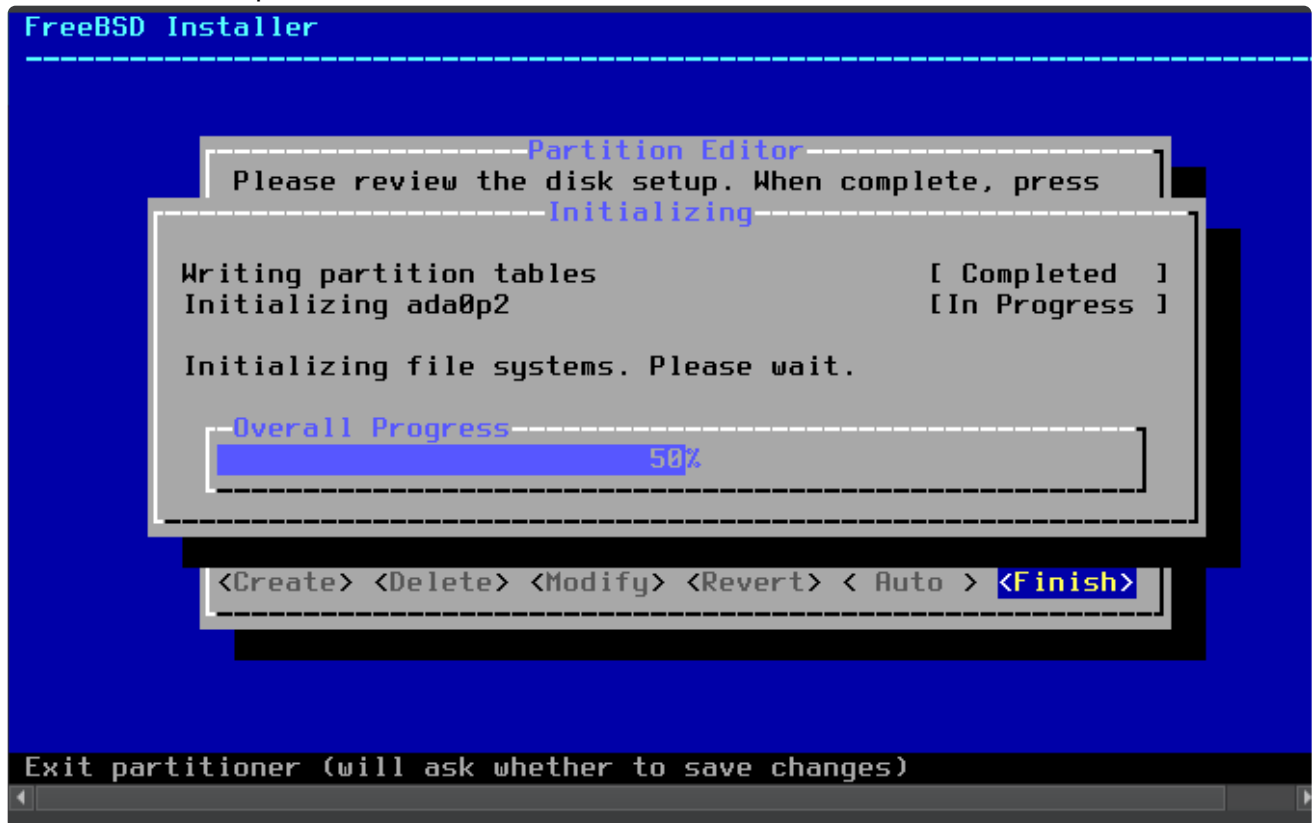
- Partition Editor, Select **Finish** and press **Enter**



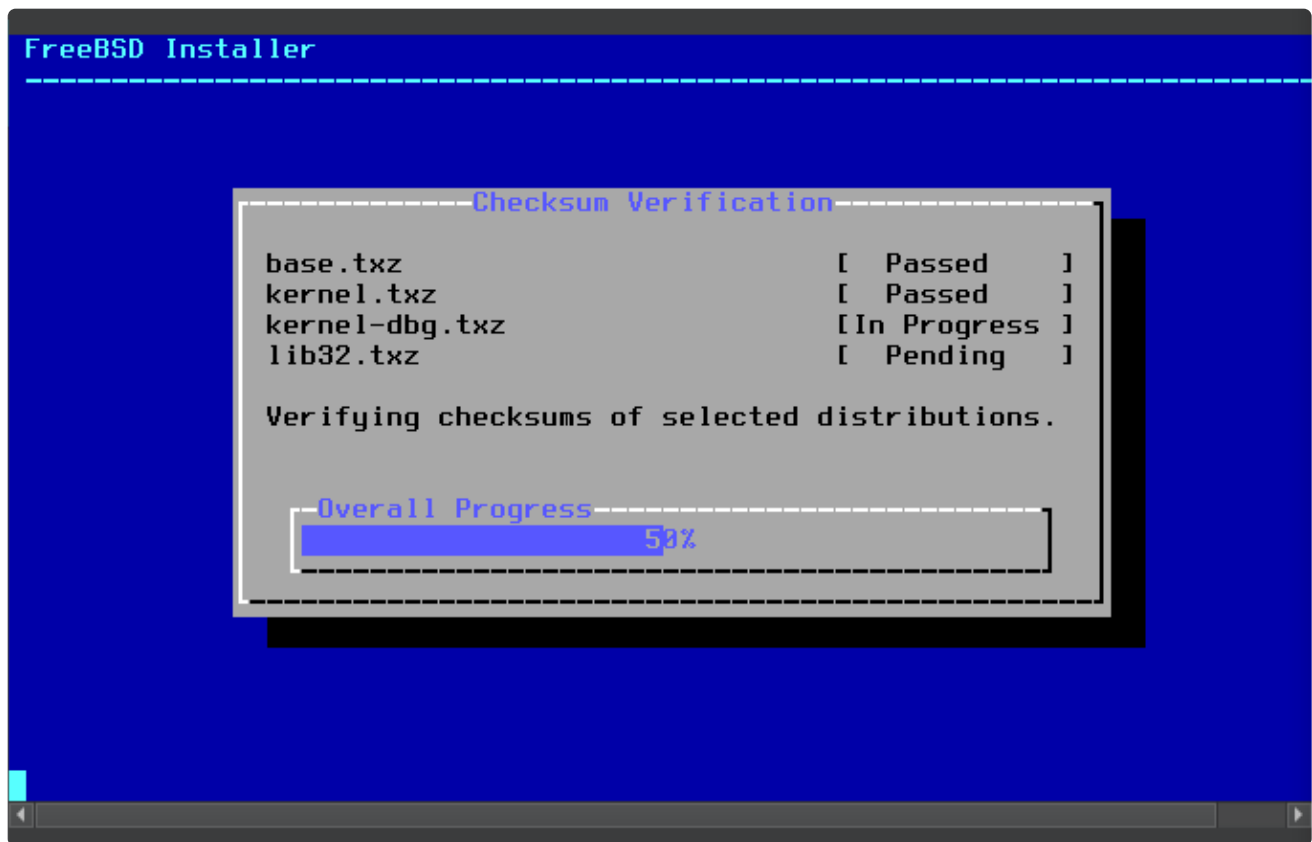
- Confirmation. Select **Commit** and press **Enter**



- Wait for it to complete



- distribution files extraction



- Set password

```
FreeBSD Installer
=====

Please select a password for the system management account (root):
Typed characters will not be visible.
Changing local password for root
New Password:█
```

- VM1 root password : `root`
- Select Network Configuration select `OK` and press `Enter`

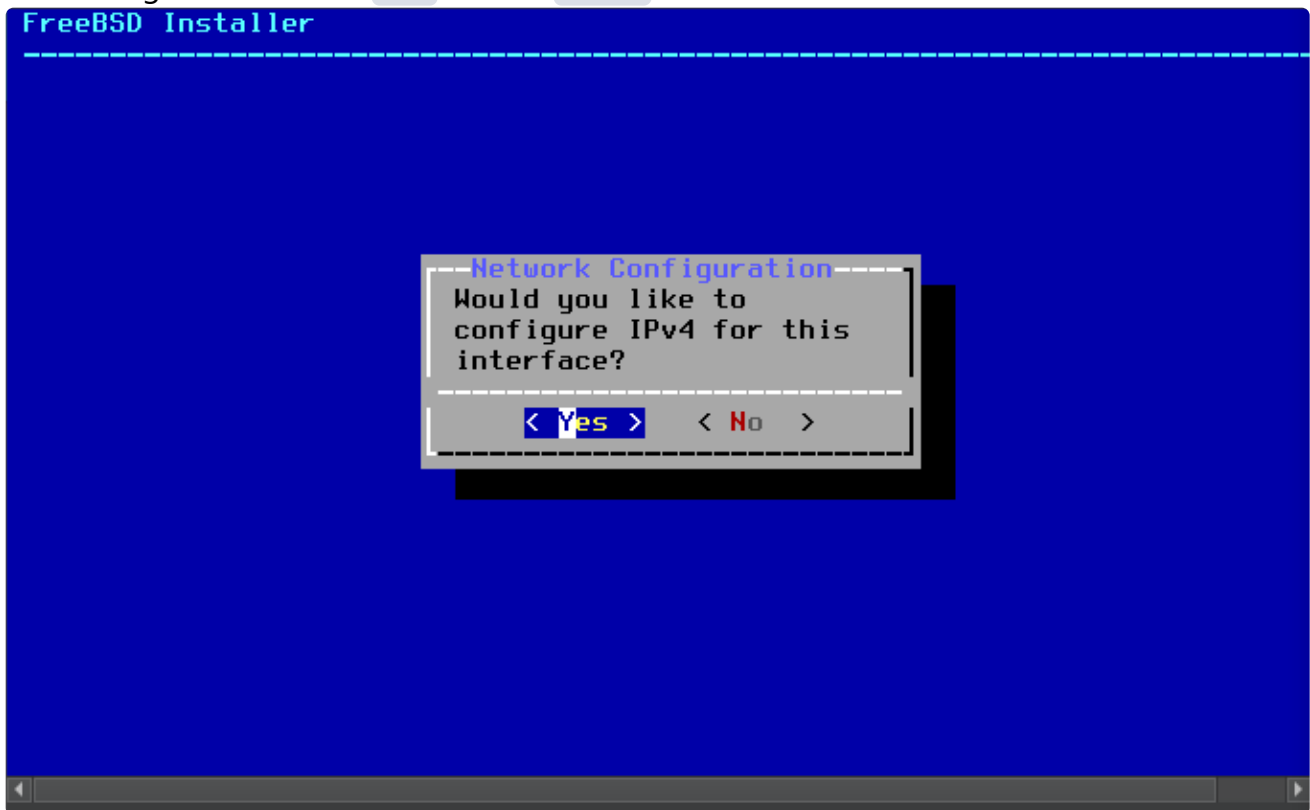
```
FreeBSD Installer
-----

Network Configuration
Please select a network interface to configure:

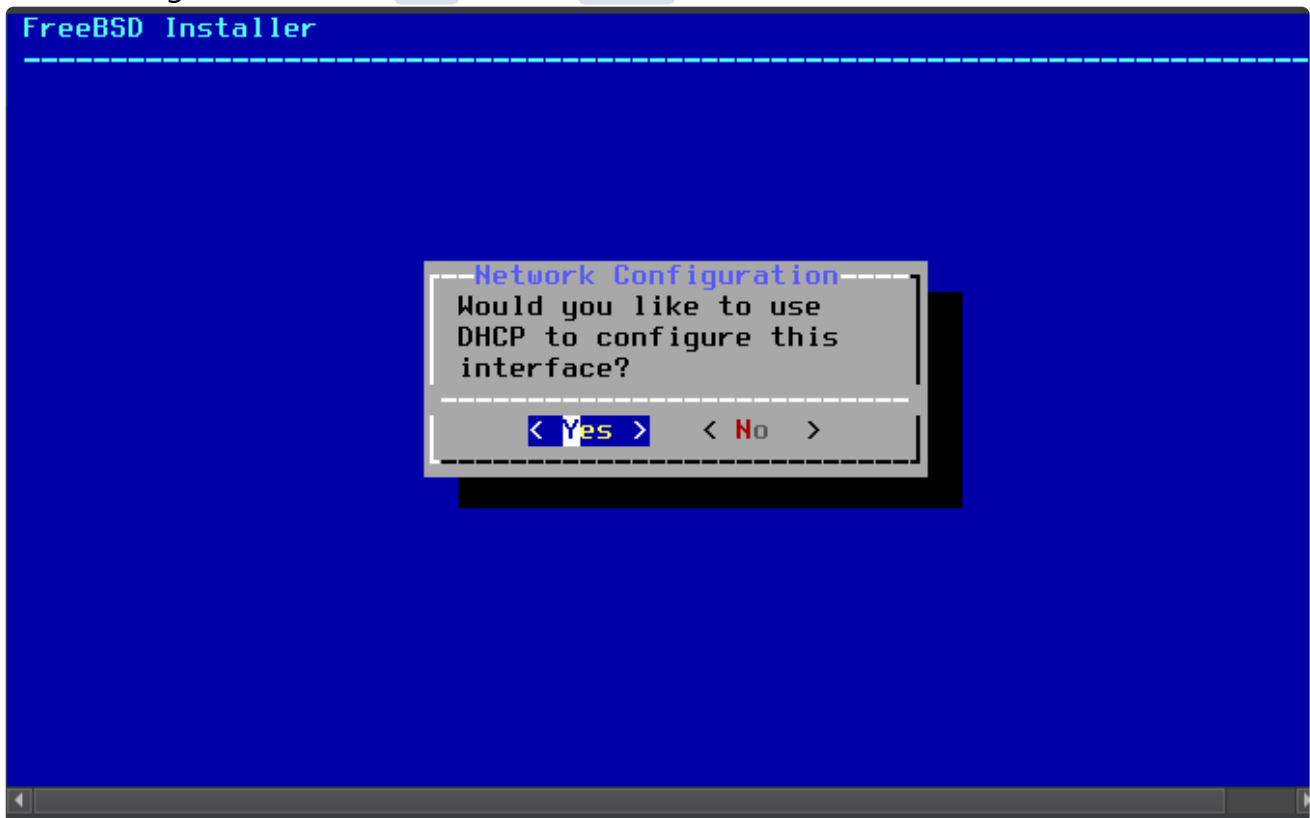
em0 Intel(R) Legacy PRO/1000 MT 82540EM

< OK >      <Cancel>
```

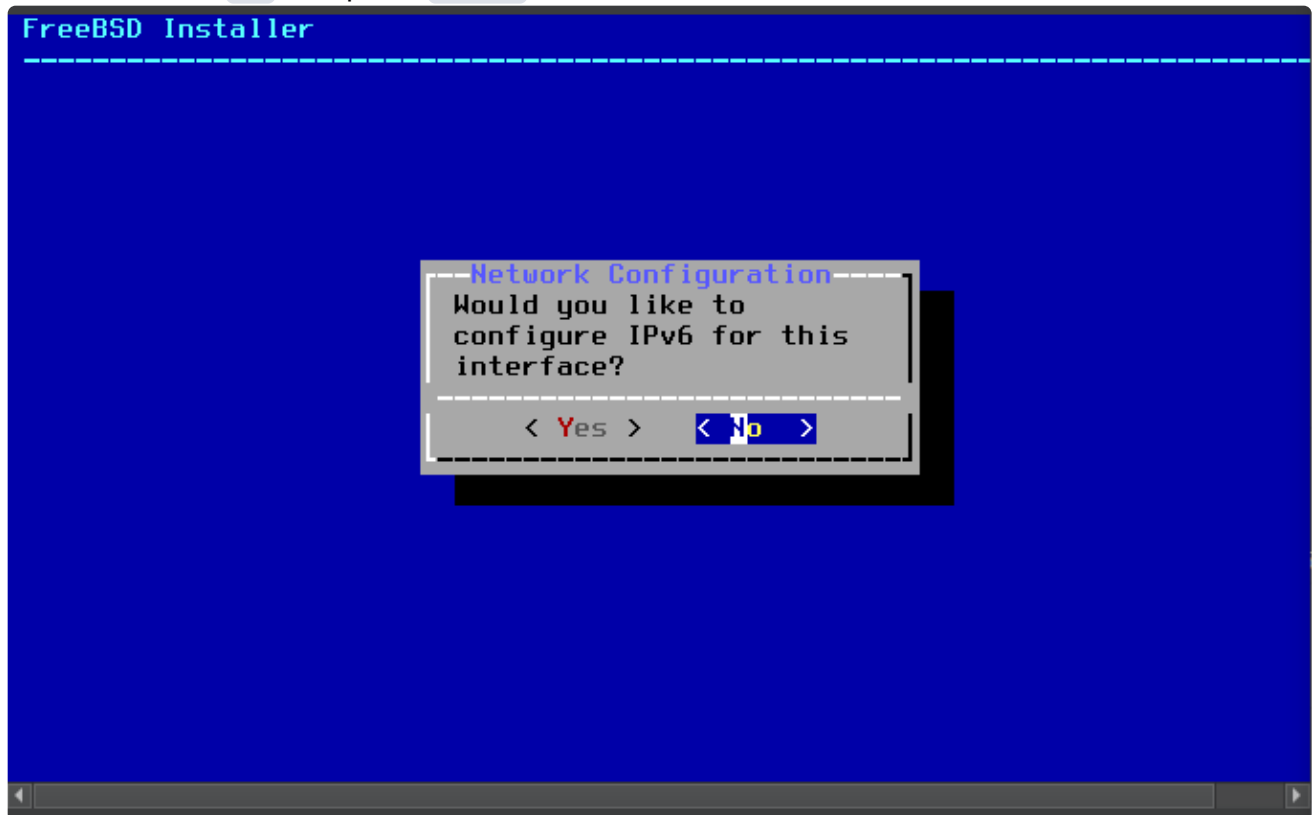
- IPv4 configuration. Select **Yes** and hit **Enter**



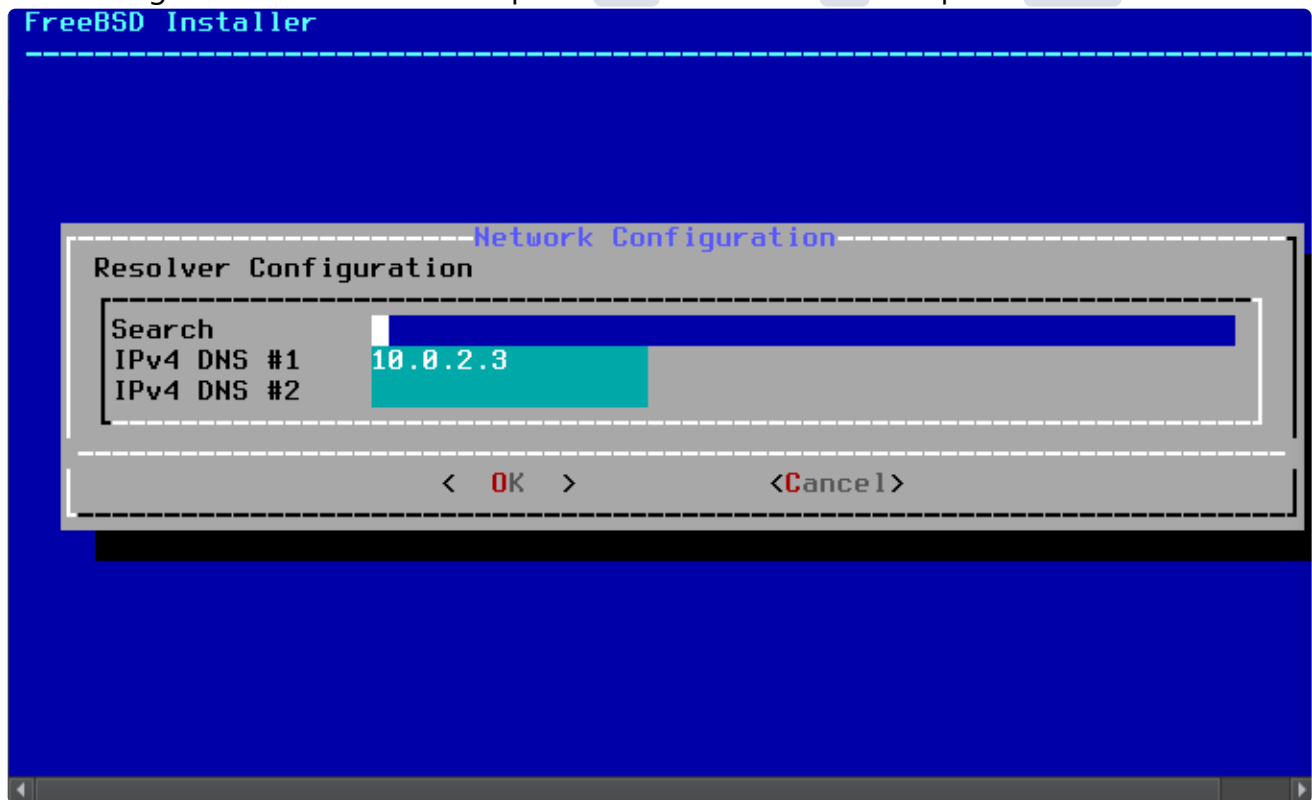
- DHCP configuration. Select **Yes** and hit **Enter**



- For IPv6, select **No** and press **Enter**



- DNS configuration. leave as default press **TAB** and select **OK** and press **Enter**



- Setup Timezone.



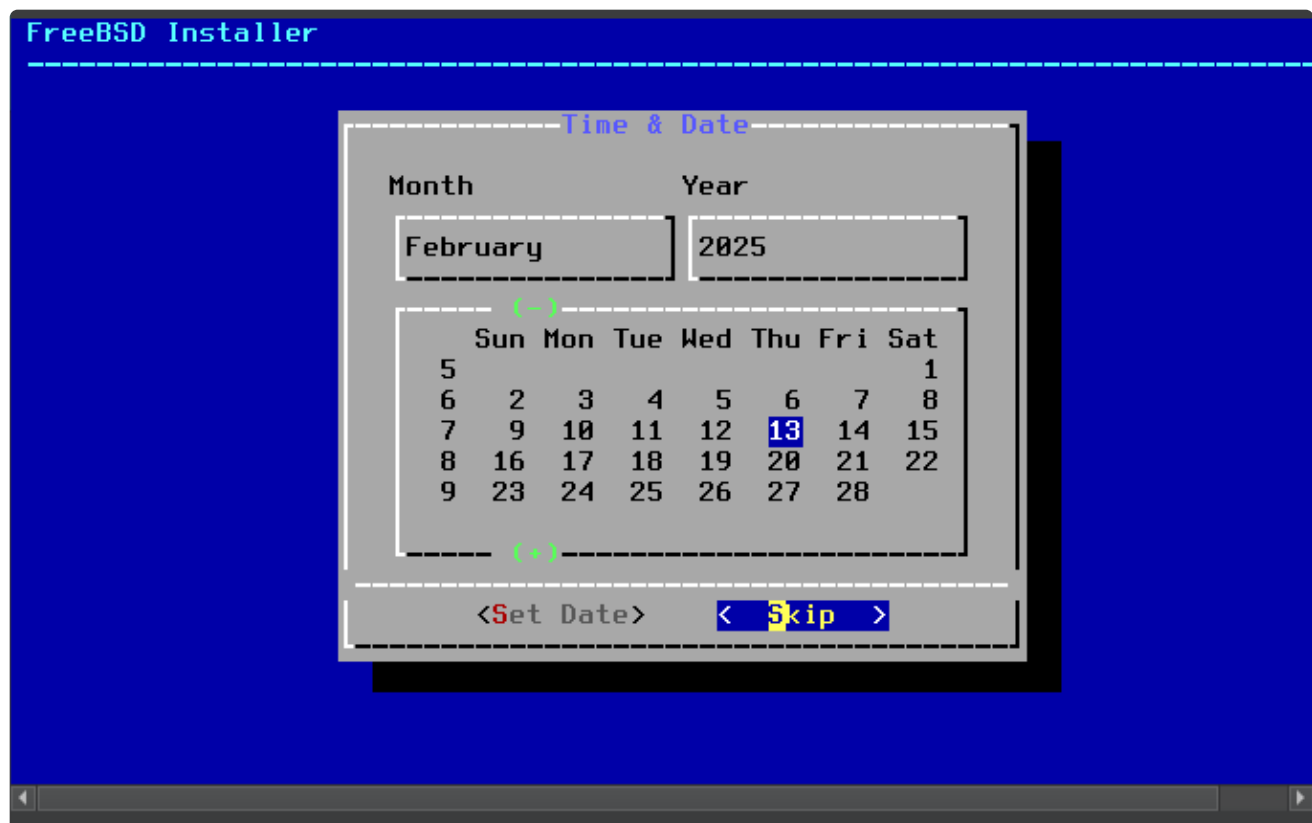
- Country



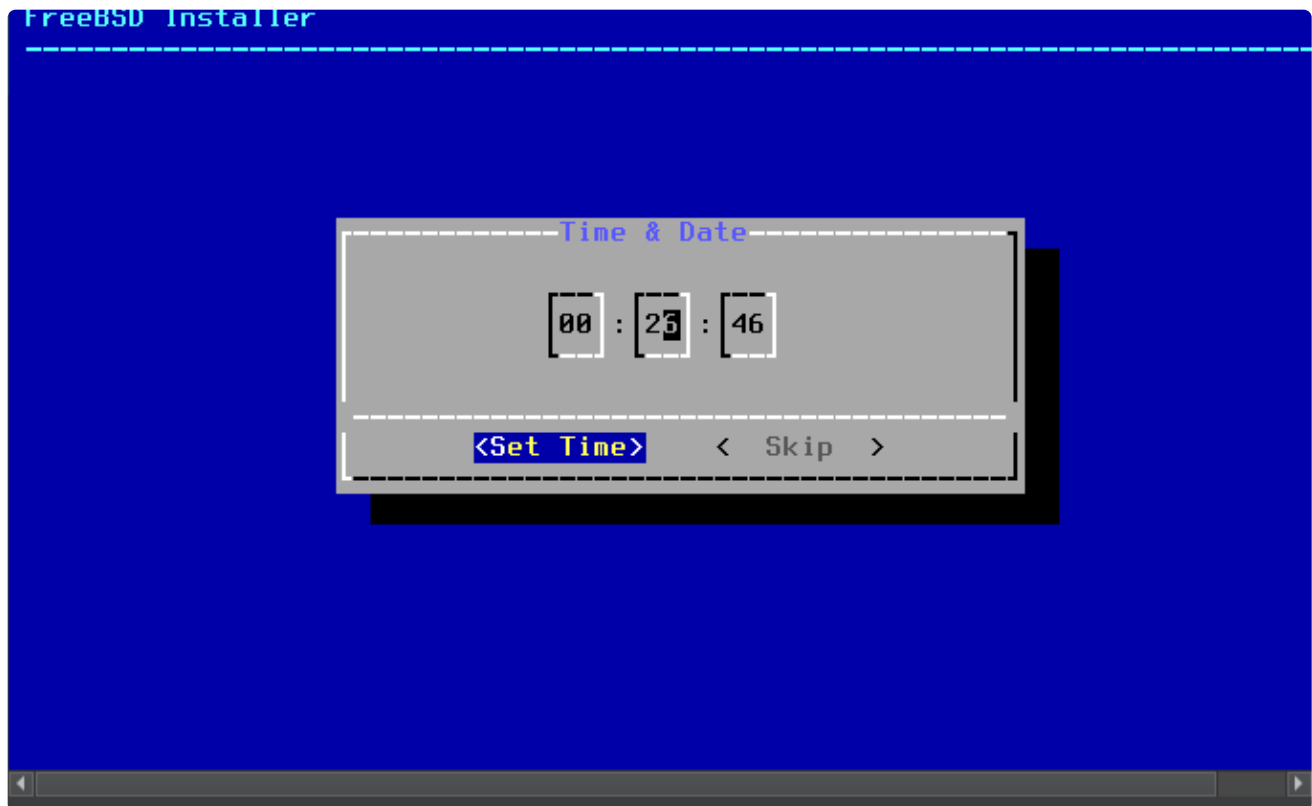
- Abbreviation



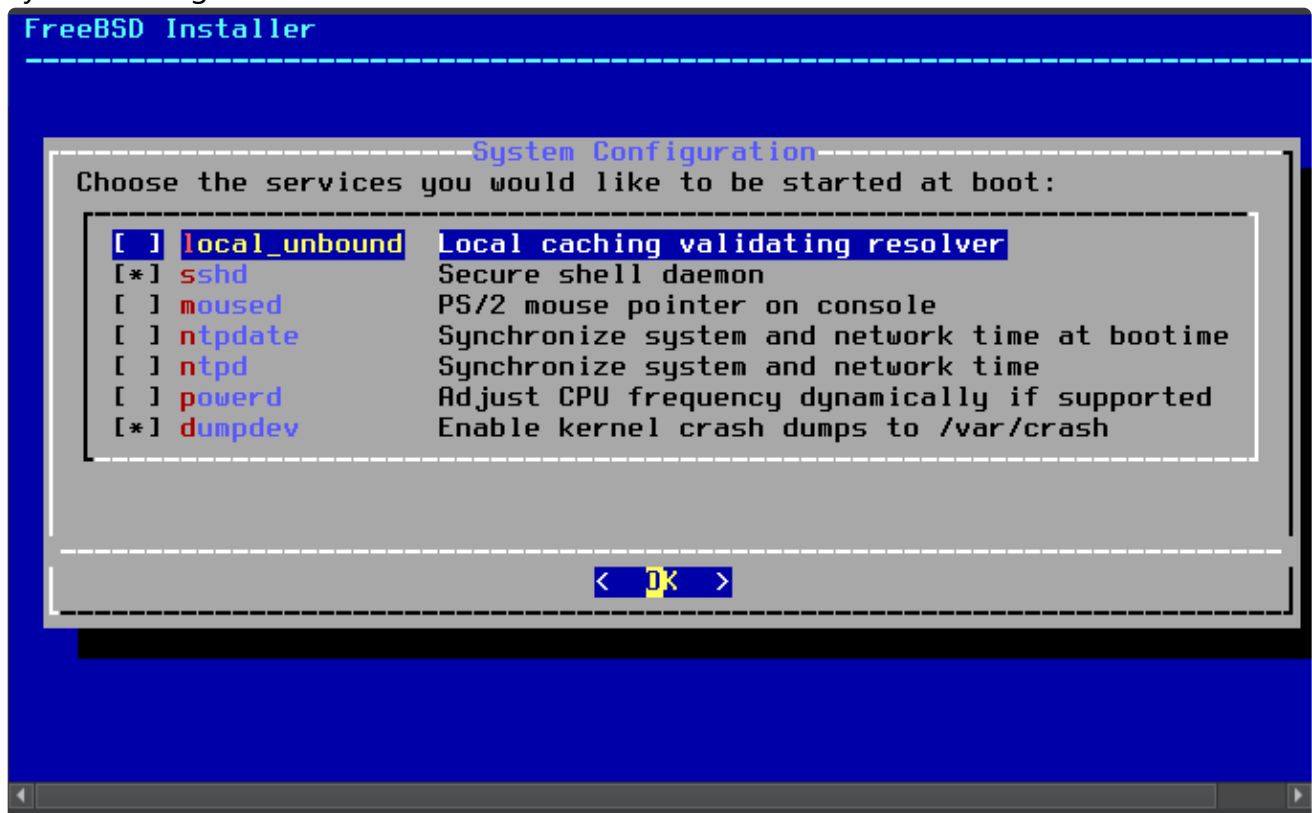
- Date



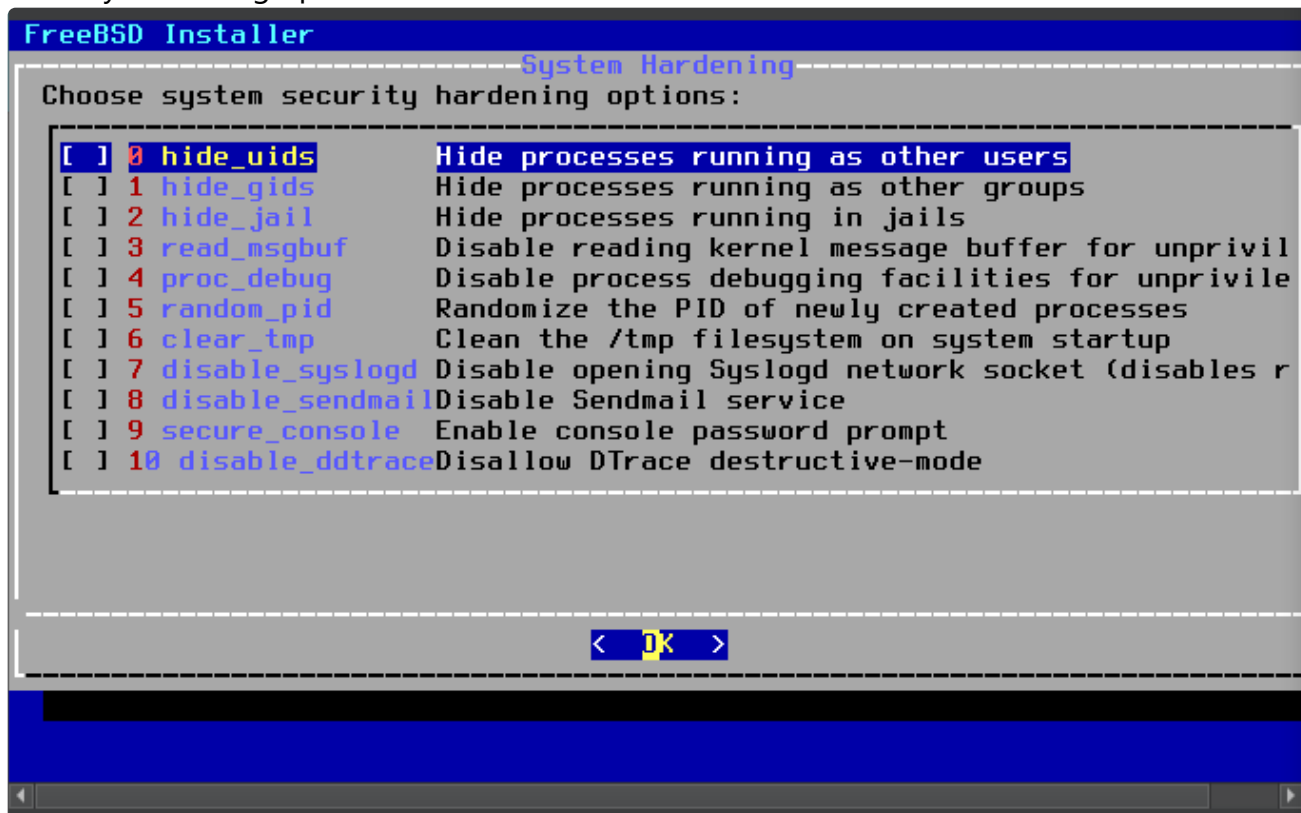
- Time



- System Configuration. Leave Defaults



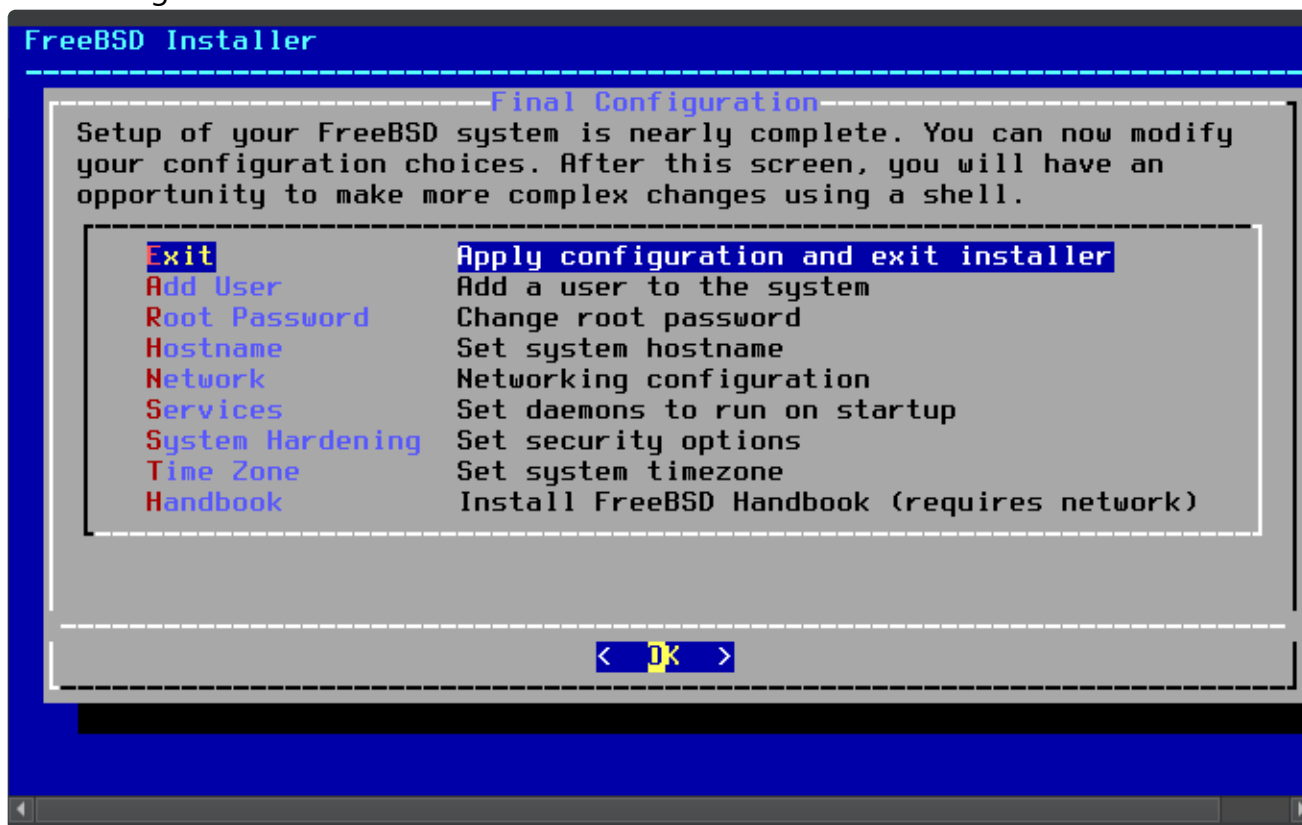
- Security hardening options. Leave Default.



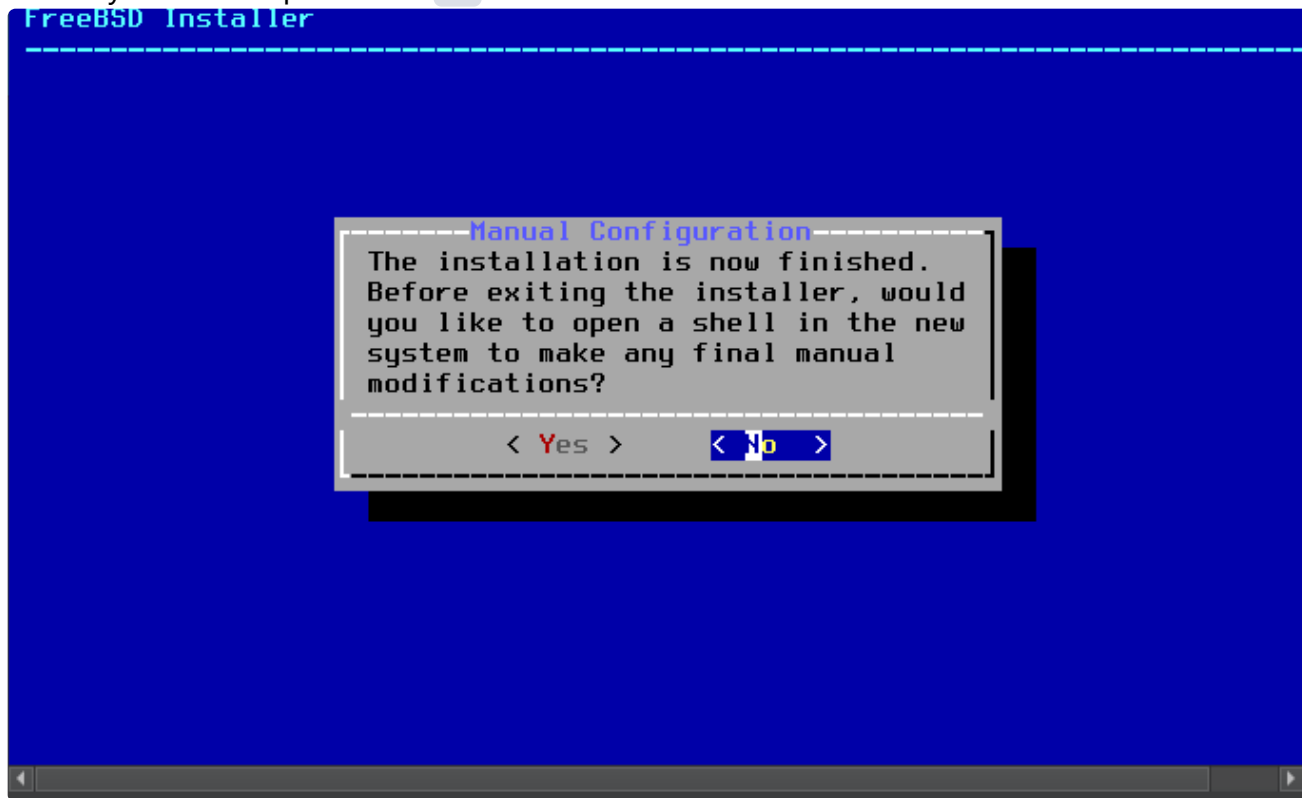
- VM user:

```
Full name: vm1
Uid (Leave empty for default):
Login group [vm1]: wheel
Login group is wheel. Invite vm into other groups? []:
Login class [default]:
Shell (sh csh tcsh nologin) [sh]:
Home directory [/home/vm]:
Home directory permissions (Leave empty for default):
Use password-based authentication? [yes]:
Use an empty password? (yes/no) [no]:
Use a random password? (yes/no) [no]:
Enter password:
Enter password again:
Lock out the account after creation? [no]:
Username      : vm
Password      : *****
Full Name     : vm1
Uid           : 1001
Class         :
Groups        : wheel
Home          : /home/vm
Home Mode     :
Shell         : /bin/sh
Locked        : no
OK? (yes/no) [yes]:
```

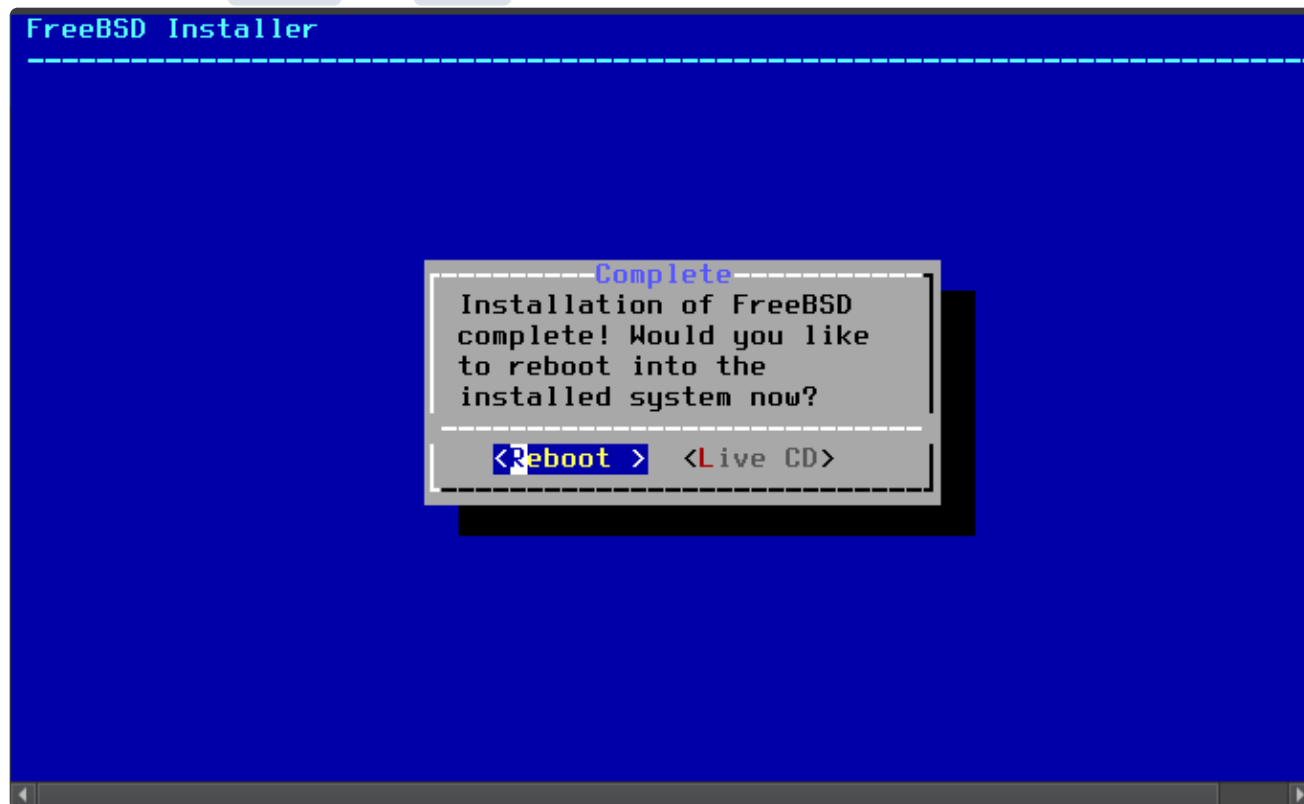
- Final configuration. Leave defaults.



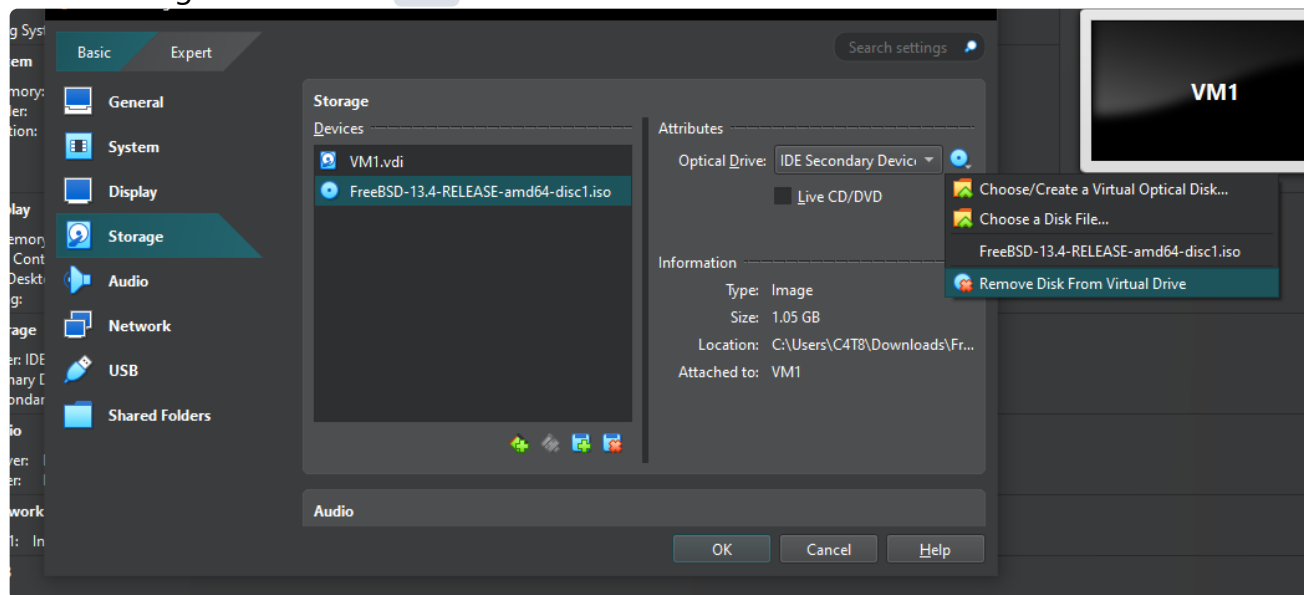
- Would you like to open shell? NO



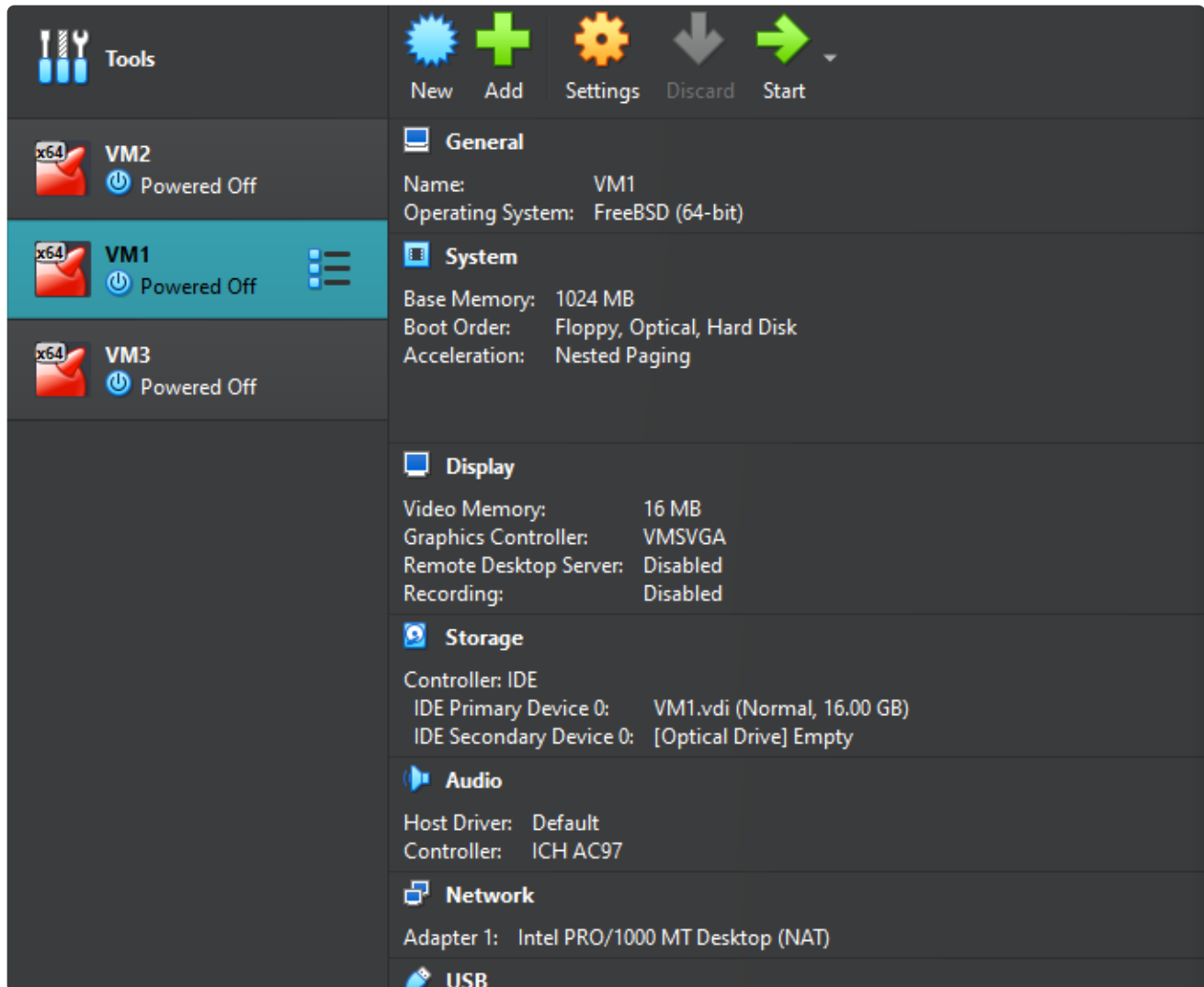
- Reboot? Select **Reboot** and **Enter**



- Close VM.
- Go to settings and remove **iso** file from devices.



- Clone VM1 to VM3



What is L3 forwarding firewall?

- An "L3 forwarding firewall" refers to a firewall that operates at the network layer (Layer 3) of the OSI model, making decisions about network traffic based solely on the source and destination IP addresses in the packet, essentially controlling which packets can be routed to different networks based on their IP addresses, without inspecting the content of the packet itself; it primarily functions as a routing device with added security features to filter traffic based on IP addresses and protocols

pf.conf file

- There is no pf.conf file in a default FreeBSD installation.

- Copy `/usr/share/examples/pf/pf.conf` file to `/etc/pf.conf`

```
vansil@firewall:/etc $ cd /usr/share/examples/pf/
vansil@firewall:/usr/share/examples/pf $ ls
ackpri      faq-example2  pf.conf      queue2       queue4
faq-example1 faq-example3  queue1       queue3       spamd
vansil@firewall:/usr/share/examples/pf $ cp pf.conf /etc/pf.conf
cp: /etc/pf.conf: Permission denied
vansil@firewall:/usr/share/examples/pf $ sudo cp pf.conf /etc/pf.conf
Password:
vansil@firewall:/usr/share/examples/pf $
```

- enabling PF with no rule set or with `pf.conf` content that is syntactically invalid will leave the packet filter enabled with a default pass all rule set.

How to create rule set

pf.conf file

block in all

pass out all keep state

- This rule set denies all incoming traffic, allows traffic we send, and retains state information on our connections. PF reads rules from top to bottom; the last rule in a rule set that matches a packet or connection is the one that is applied

Enable IP Routing in FreeBSD virtual machine

To enable IP routing on a FreeBSD virtual machine, you need to set the `sysctl` variable `net.inet.ip.forwarding` to 1 using the command: `sysctl net.inet.ip.forwarding=1`; this allows the system to forward packets between different networks.

Explanation:

- **Accessing the system settings:** The `sysctl` command is used to access and modify system settings at runtime.
- `net.inet.ip.forwarding` variable: This specific variable controls whether the system is allowed to forward IP packets.
- **Setting to 1:** By setting the value to 1, you enable IP routing functionality.

Important Considerations:

- **Persistence:** To make the change permanent across reboots, you can add the following line to your `/etc/sysctl.conf` file:

```
net.inet.ip.forwarding=1
```

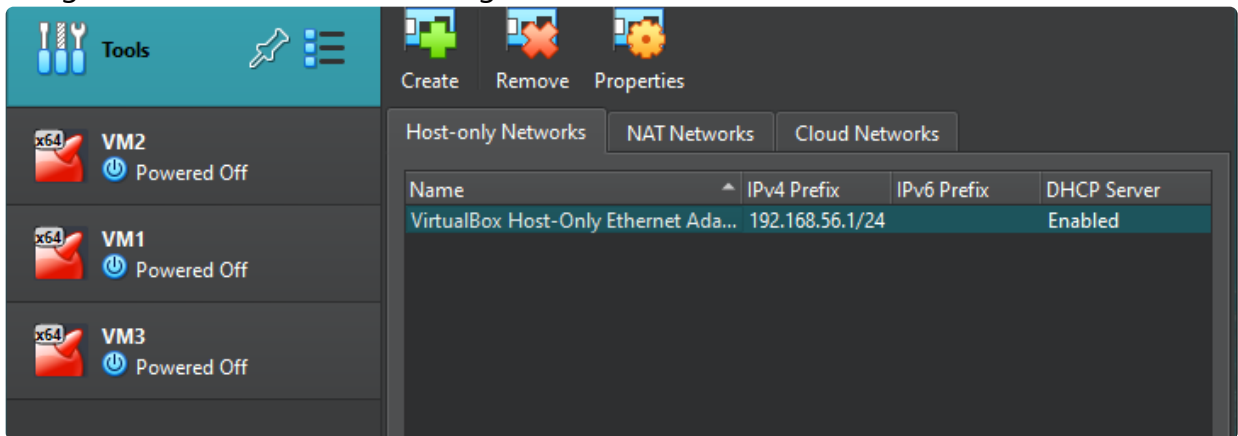
- **Firewall configuration:** Ensure your firewall rules are configured appropriately to allow traffic to be forwarded as needed.

How to connect VMs using virtual network.

To connect Virtual Machines (VMs) using a virtual network in VirtualBox, you need to **configure each VM to use a "Host-only Adapter" network**, which allows them to communicate directly with each other within the VirtualBox environment; essentially creating a private network between the VMs by setting the network adapter in each VM to use the same "Host-only Adapter" network within VirtualBox settings.

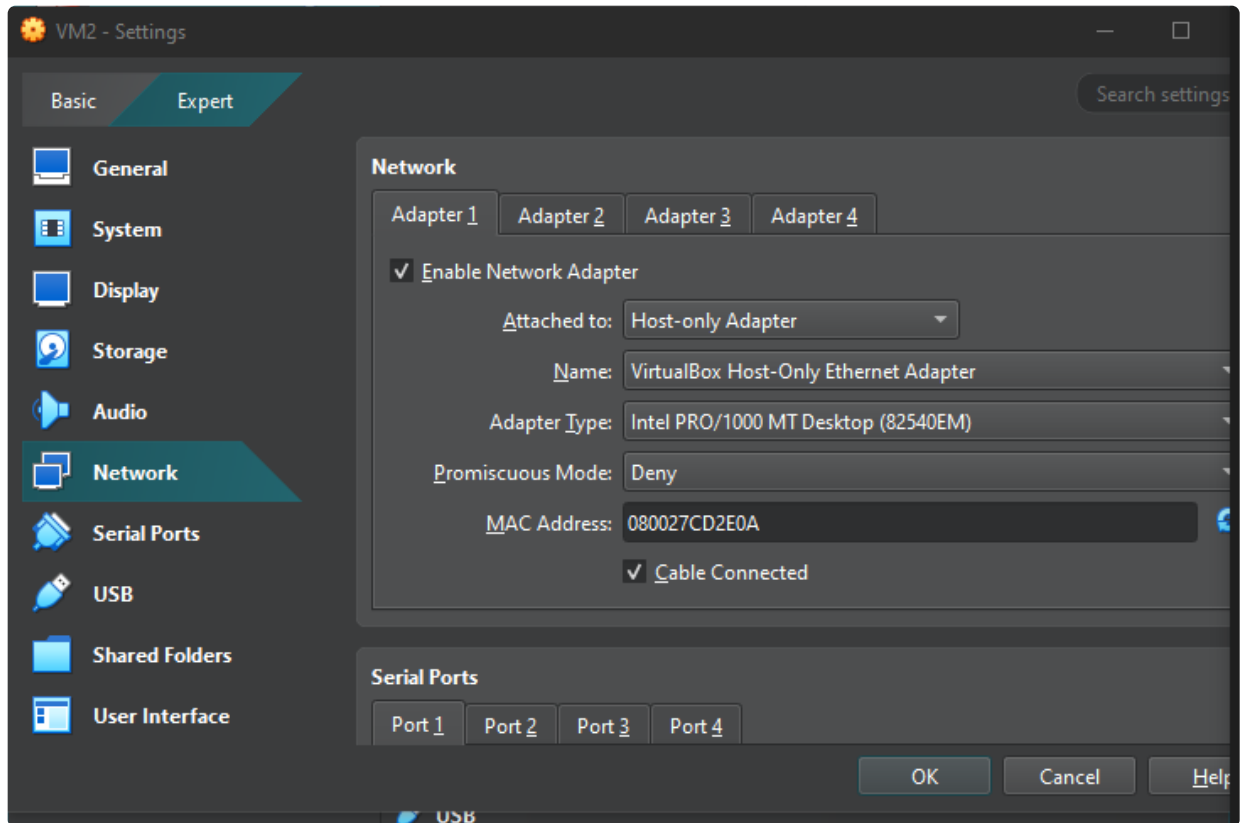
Key steps:

- **Create a Host-only Network:**
 - Open VirtualBox settings (File > Preferences).
 - Go to the "Network" tab.
 - Click the "Add" button to create a new Host-only Network.
 - Assign a name for the network (e.g., "vboxnet0").

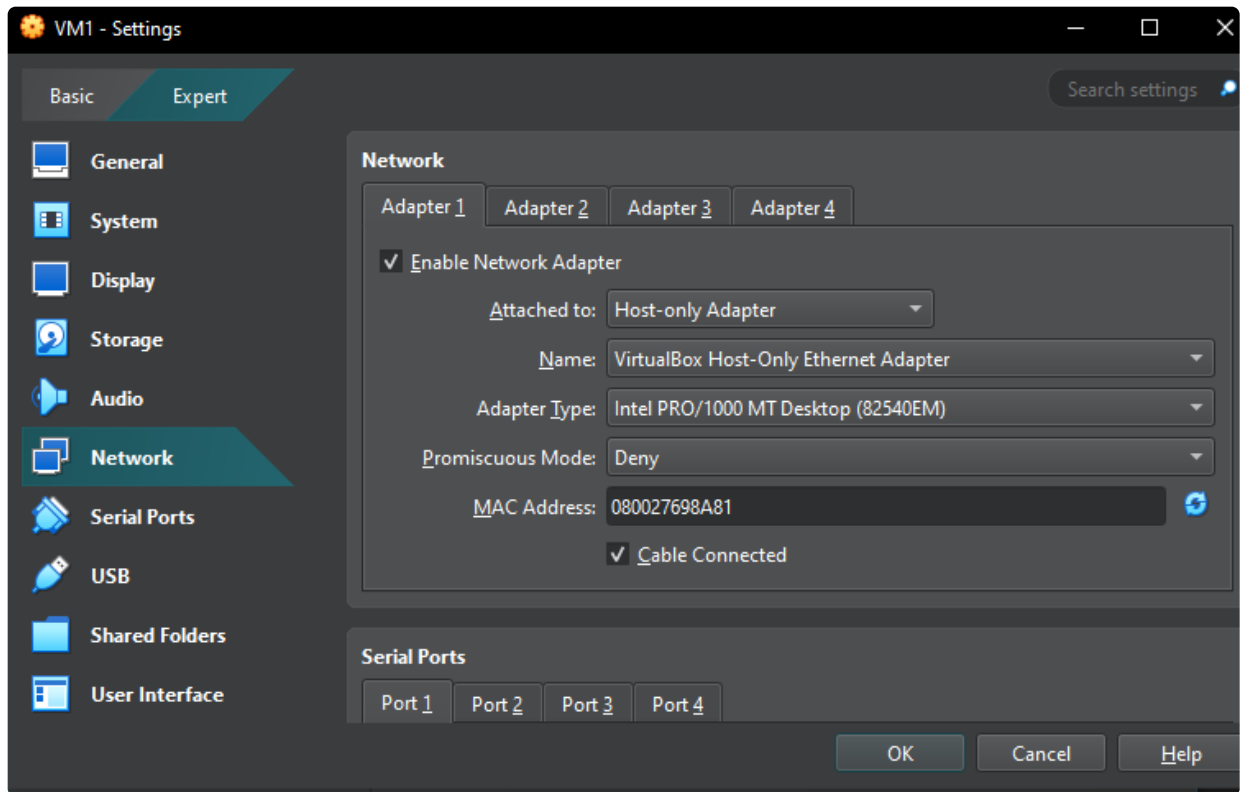


- **Configure VM Network Settings:**
 - Open the VM settings for each machine you want to connect.
 - Go to the "Network" tab.
 - Select "Attached to:" as "Host-only Adapter".
 - Choose the Host-only network you created from the dropdown menu.

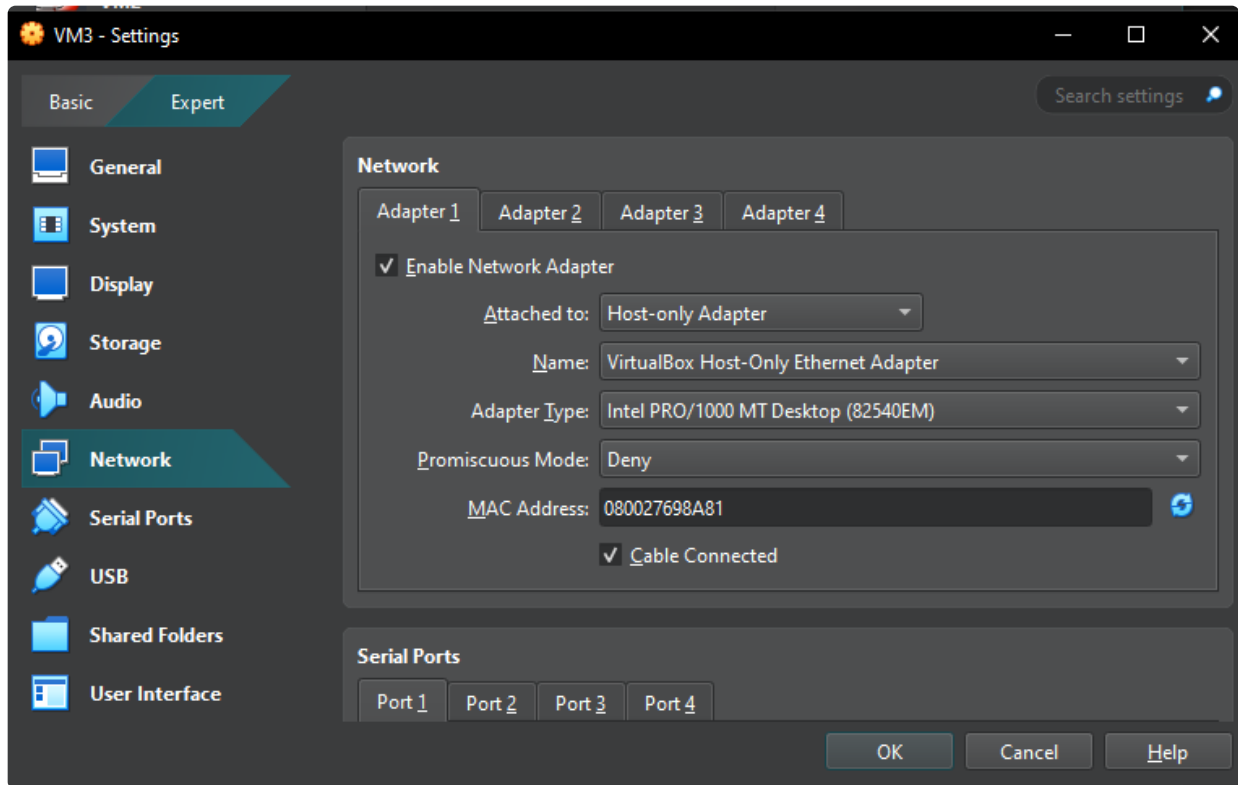
- VM2



- VM1:



- VM2:



- **Assign IP addresses:**
 - Inside the guest operating system of each VM, access network settings.
 - Set static IP addresses within the same subnet for each VM, ensuring they can reach each other.

Important points to remember:

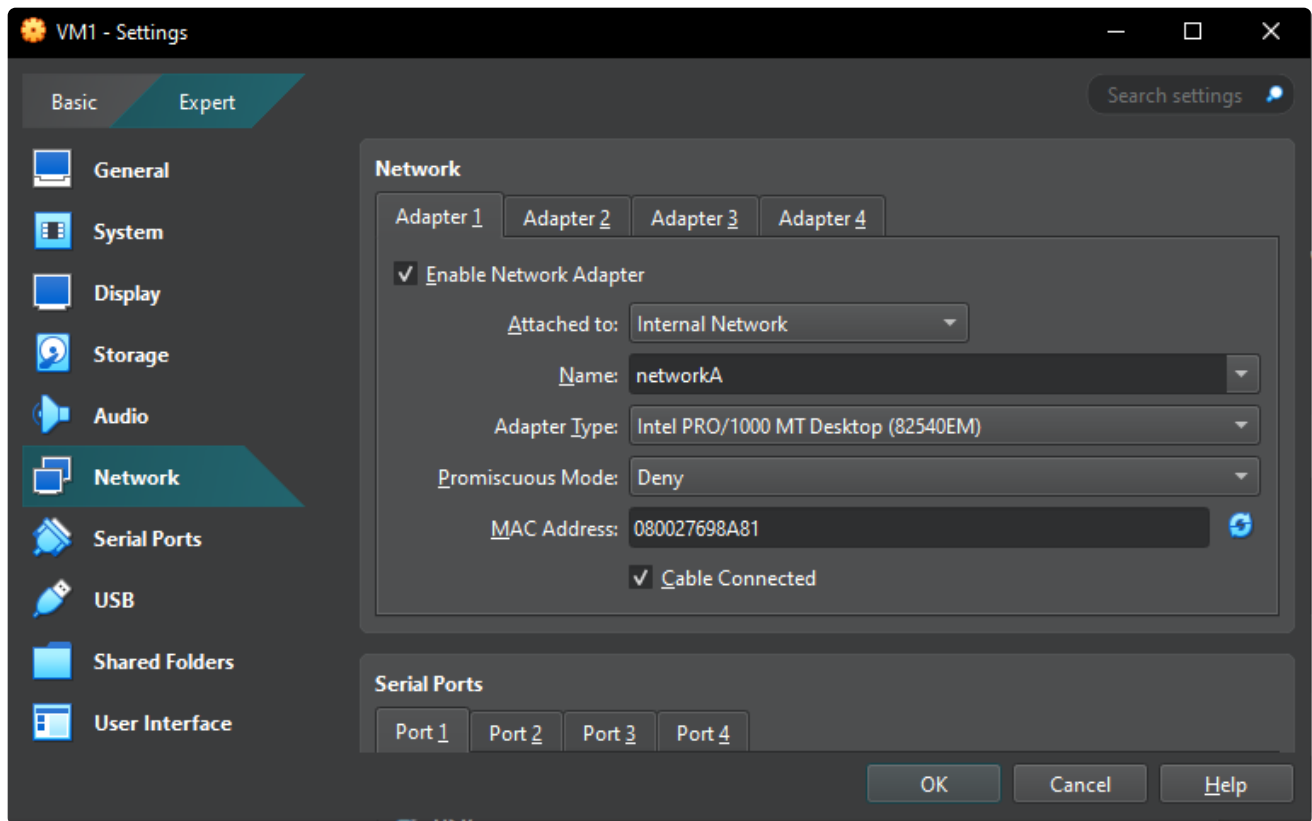
- **DHCP server:** You can enable a DHCP server on the Host-only network to automatically assign IP addresses to VMs.
- **Network connectivity test:** Once configured, use the "ping" command within each VM to verify connectivity with the other VMs.

All VMs now in one network. Ping each other

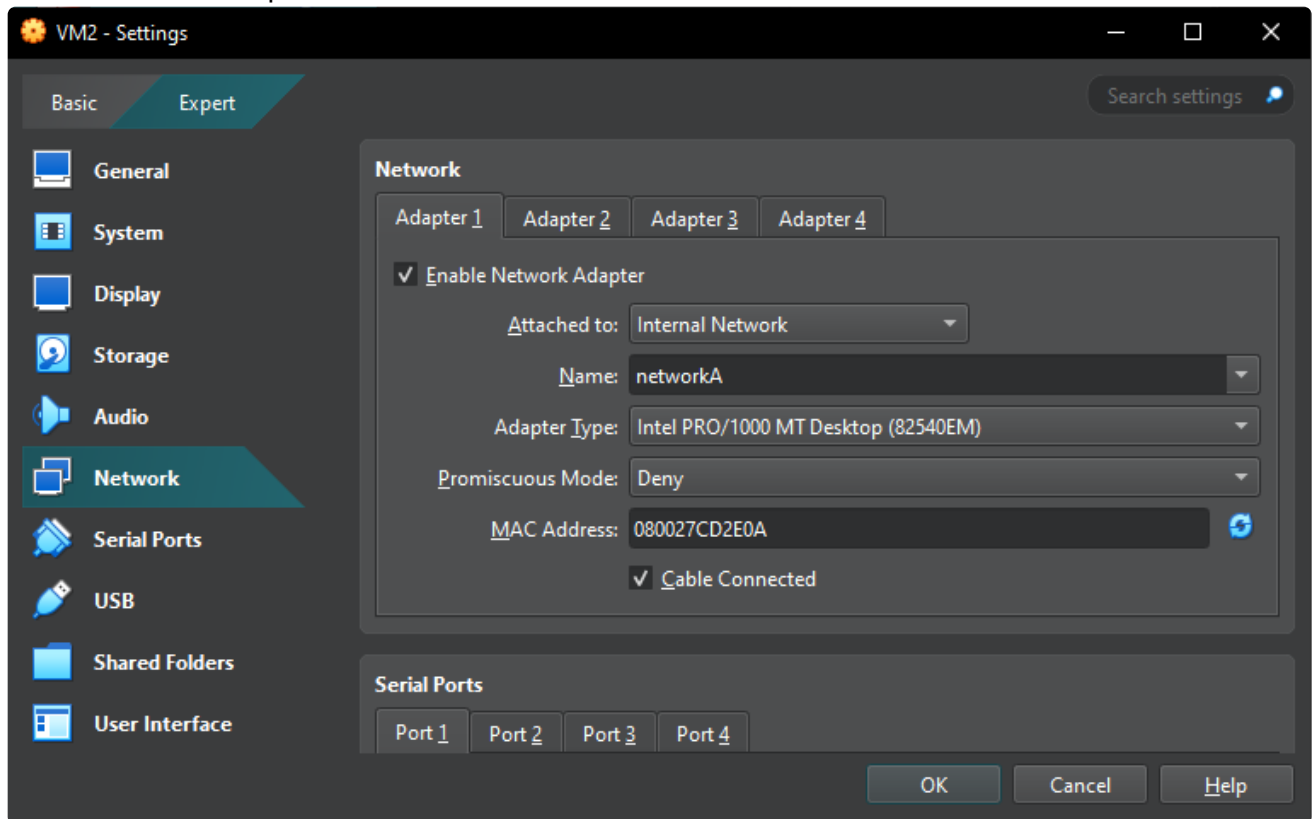
Create 2 subnets and 2 adapters in VM2.

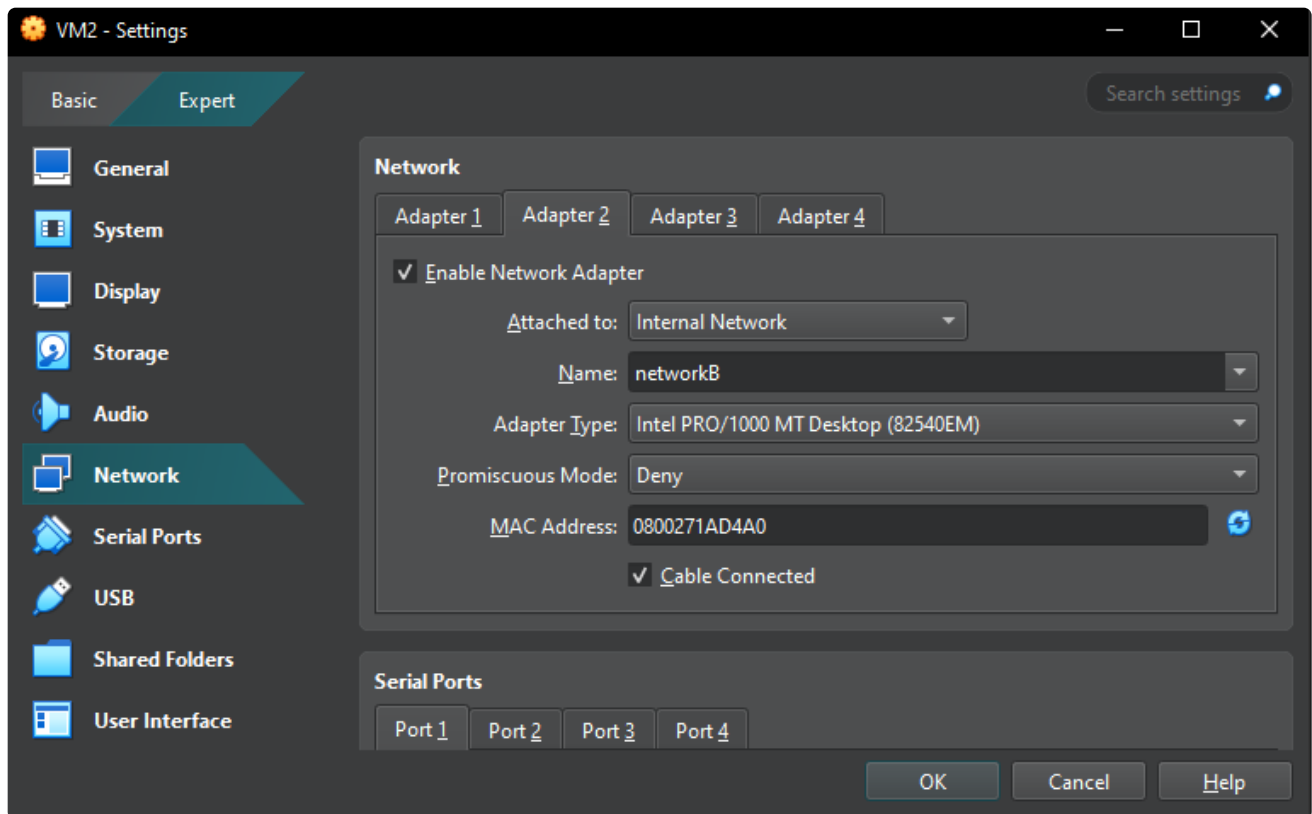
- We need to create two internal networks.
- Add appropriate settings for each VMs

- For VM1:

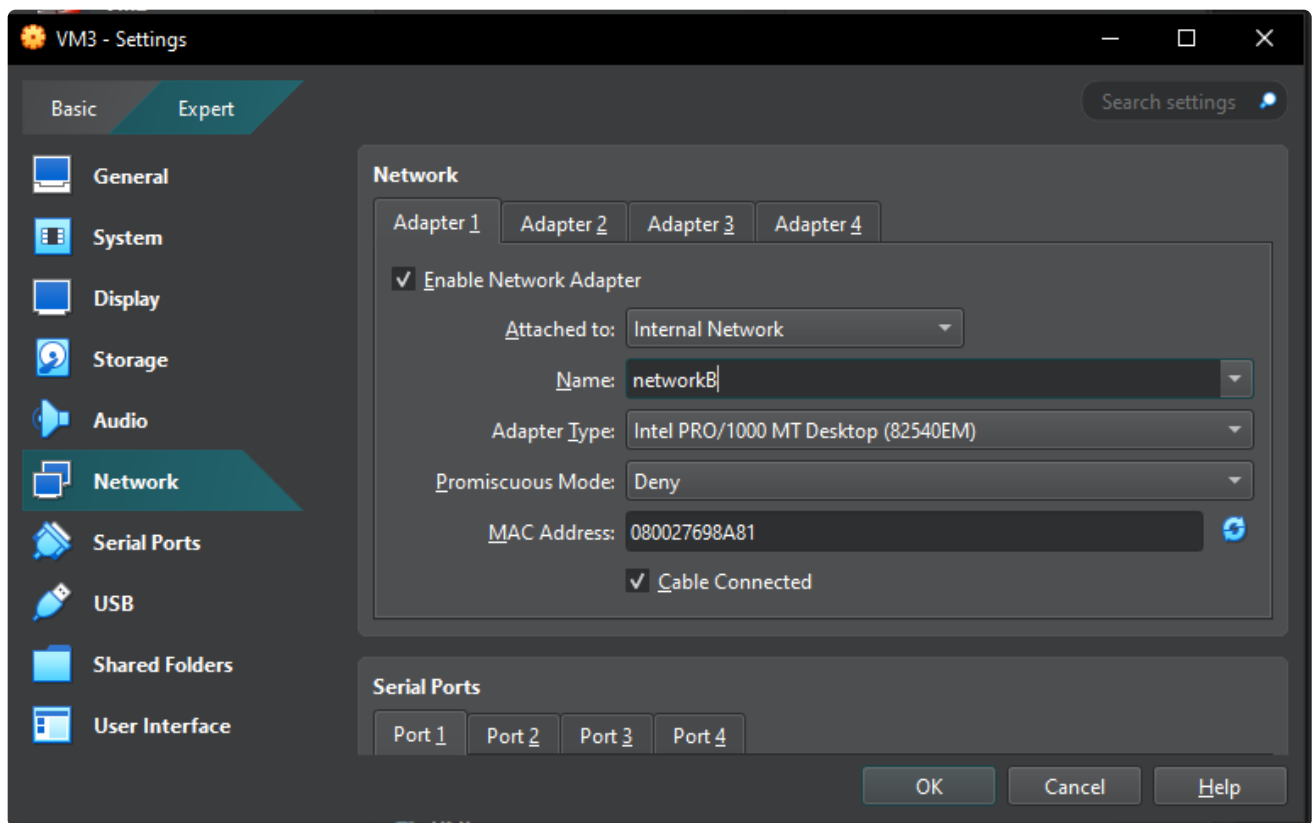


- For VM2 (Two Adapters):





- VM3:



- VM1 (adapter 1) and VM2 (adapter 1) = networkA
- VM2 (adapter 2) and VM3 (adapter 1) = networkB

How to Set IP address

- Let's first only communicate with VM1 and VM2 using ping.
- Connected Both `VM1` and `VM2` using internal network `networkA`
- Setting up IP address in both
- Setup IP address command.

Identify your network interface:

- Use the command `ifconfig -a` to list all network interfaces and their current status.

```
vm@vm1:~ $ ifconfig -a
em0: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
    options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMBO_MTU,VLAN
SUM,TSO4,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
    ether 08:00:27:69:8a:81
    inet 0.0.0.0 netmask 0xff000000 broadcast 255.255.255.255
    media: Ethernet autoselect (1000baseT <full-duplex>)
    status: active
    nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
    options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_IPV6>
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
    inet 127.0.0.1 netmask 0xff000000
    groups: lo
    nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
vm@vm1:~ $
```

- Note the name of the interface you want to configure (e.g., `em0`, `re0`, `vtnet0`).
 - `em0`

Open the `/etc/rc.conf` file:

- Use a text editor (e.g., `nano`, `vi`) to open the file: `sudo nano /etc/rc.conf`.

1. Configure the network settings:

- Set the hostname: `hostname="your_hostname"`.
- Set the IP address, netmask, and gateway: For example:
 - `ifconfig_em0="inet 192.168.1.100 netmask 255.255.255.0"`.
 - `defaultrouter="192.168.1.1"`.

Enable SSH in FreeBSD

- Login with `root`

```
pkg update
pkg upgrade
```

- Enable SSH service

```
sudo nano /etc/rc.conf
```

- Add following line to the file:

```
sshd_enable="YES"
```

- Exit editor and start ssh service

```
sudo service sshd start
```

- Change adapter to `Bridged Adapter`
- Try using ssh from terminal

```
find / -name "*GENERIC*" -ls

will search '/', and all subdirectories, for files with 'GENERIC'
-- Stephen Hilton <nospam@hiltonbsd.com>
vansil@firewall:~ $ ls
temp.txt
vansil@firewall:~ $ rm temp.txt
vansil@firewall:~ $ ls
vansil@firewall:~ $
```

Packet Filtering

- PF is a stateful firewall by default
- Pf filters packets according to three core actions:
 - `block`, `pass` and `match`
 - An action is taken when a packet meets the criteria specified in a rule.

Edit `rc.conf` to enable `PF`

```
sudo sysrc pf_enable="YES"  
sudo sysrc pflog_enable="YES"
```

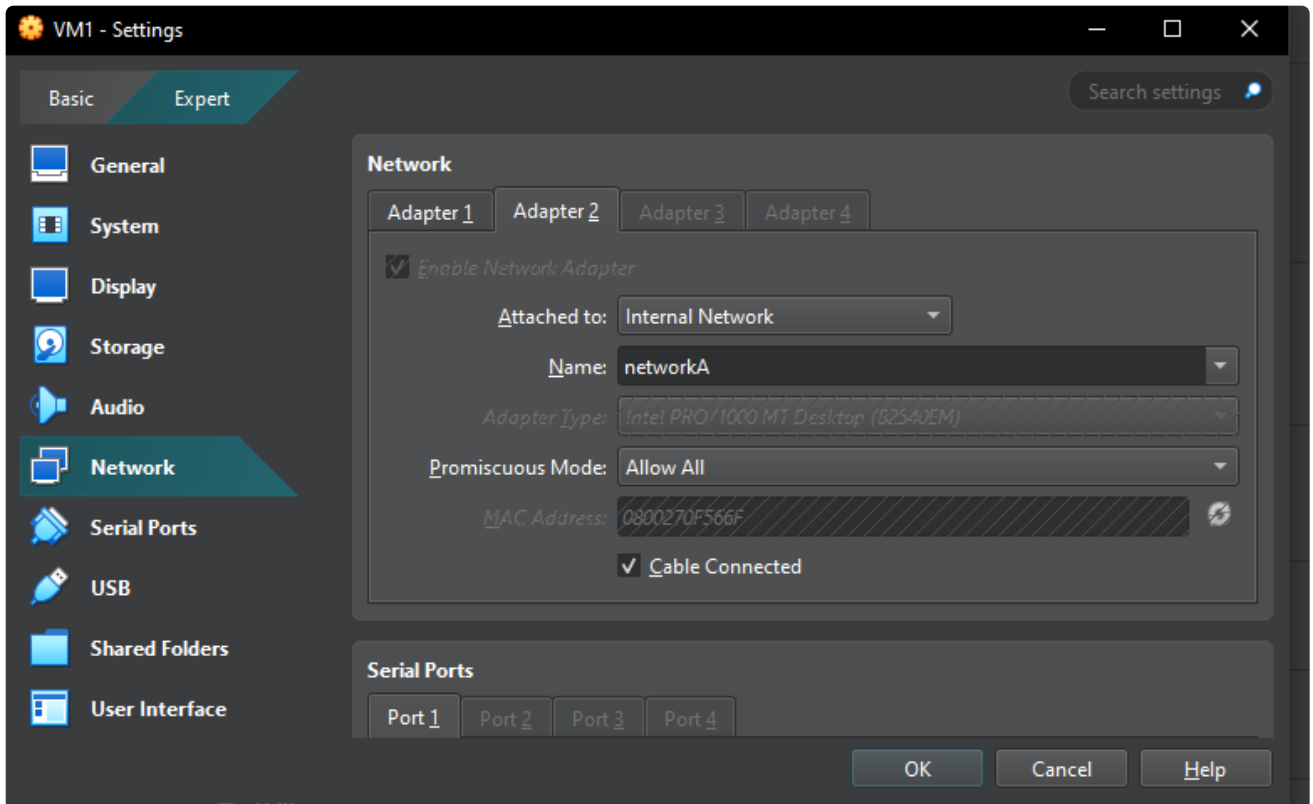
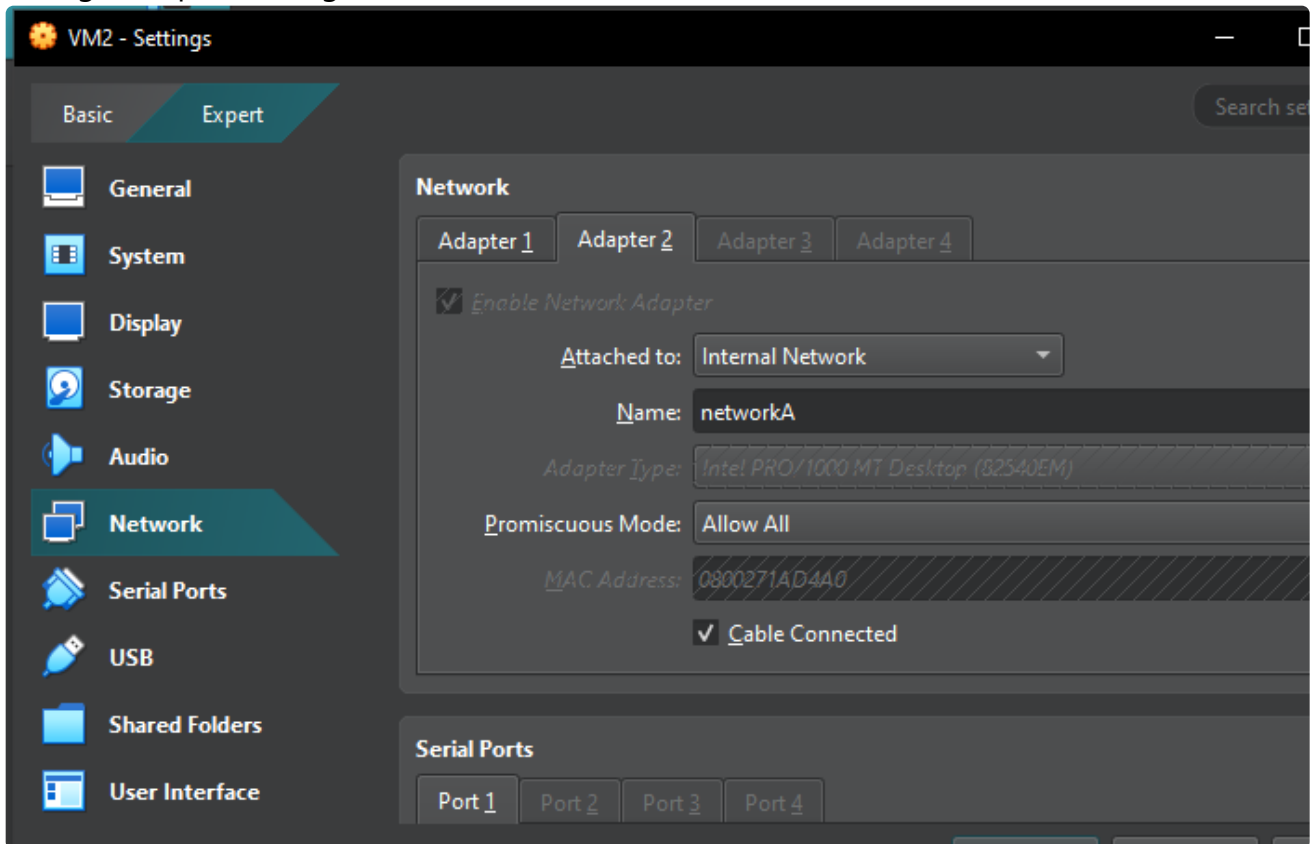
Edit `pf.conf` file

```
set skip on lo0  
block all  
pass in proto tcp to port { 22 }  
pass out proto { tcp udp } to port { 22 53 80 123 443 }  
pass out inet proto icmp icmp-type { echoreq }
```

```
vansil@firewall:~ $ ls  
vansil@firewall:~ $ sudo cat /etc/pf.conf  
set skip on lo0  
block all  
pass in proto tcp to port { 22 }  
pass out proto { tcp udp } to port { 22 53 80 123 443 }  
pass out inet proto icmp icmp-type { echoreq }  
vansil@firewall:~ $ |
```

Connecting two VMs (VM1 and VM2) using internal network.

- Change Adapter settings in both VMs



- Update Ip addresses in `/etc/rc.conf`

```

GNU nano 8.2 /etc/rc.conf
hostname="vm1.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.0.0 netmask 255.255.255.254"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

GNU nano 8.2 /etc/rc.conf Modified
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.0.15 netmask 255.255.255.254"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

sshd_enable="YES" # enable ssh
pf_enable="YES" # enable pf (load module if required)
pflog_enable="YES" # start pflogd(8)

add host ::1: gateway lo0
add net fe80:: gateway ::1
add net ff02:: gateway ::1
add net ::ffff:0.0.0.0: gateway ::1
add net ::0.0.0.0: gateway ::1
root@vm1:/home/vm # ifconfig
em0: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMB
O_MTU, VLAN_HWCSUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, NOMAP>
ether 08:00:27:69:8a:81
inet 192.168.0.106 netmask 0xfffff00 broadcast 192.168.0.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
em1: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMB
O_MTU, VLAN_HWCSUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, NOMAP>
ether 08:00:27:0f:56:6f
inet 192.168.0.0 netmask 0xffffffe broadcast 255.255.255.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM, TXCSUM, LINKSTATE, RXCSUM_IPV6, TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@vm1:/home/vm #

add net ff02:: gateway ::1
add net ::ffff:0.0.0.0: gateway ::1
add net ::0.0.0.0: gateway ::1
vansil@firewall:~$ ifconfig
em0: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMB
O_MTU, VLAN_HWCSUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, NOMAP>
ether 08:00:27:cd:2e:0a
inet 192.168.0.104 netmask 0xfffff00 broadcast 192.168.0.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
em1: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMB
O_MTU, VLAN_HWCSUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, NOMAP>
ether 08:00:27:1a:d4:a0
inet 192.168.0.1 netmask 0xffffffe broadcast 255.255.255.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM, TXCSUM, LINKSTATE, RXCSUM_IPV6, TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
pflog0: flags=100<PROMISC> metric 0 mtu 33160
groups: pflog
vansil@firewall:~$

```

- Reset network interfaces

`/etc/rc.d/netif restart && /etc/rc.d/routing restart`

- Ping works (after disabling pf)

```

nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPv6>
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
    inet 127.0.0.1 netmask 0xff000000
    groups: lo
    nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@vm1:/home/vm # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
^C
--- 192.168.0.1 ping statistics ---
27 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
^C
--- 192.168.0.1 ping statistics ---
4 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
^Z
Suspended
root@vm1:/home/vm #
root@vm1:/home/vm # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: icmp_seq=0 ttl=64 time=4.748 ms
64 bytes from 192.168.0.1: icmp_seq=1 ttl=64 time=3.250 ms
64 bytes from 192.168.0.1: icmp_seq=2 ttl=64 time=3.861 ms
64 bytes from 192.168.0.1: icmp_seq=3 ttl=64 time=3.146 ms
^C
--- 192.168.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.146/3.751/4.748/0.637 ms
root@vm1:/home/vm #

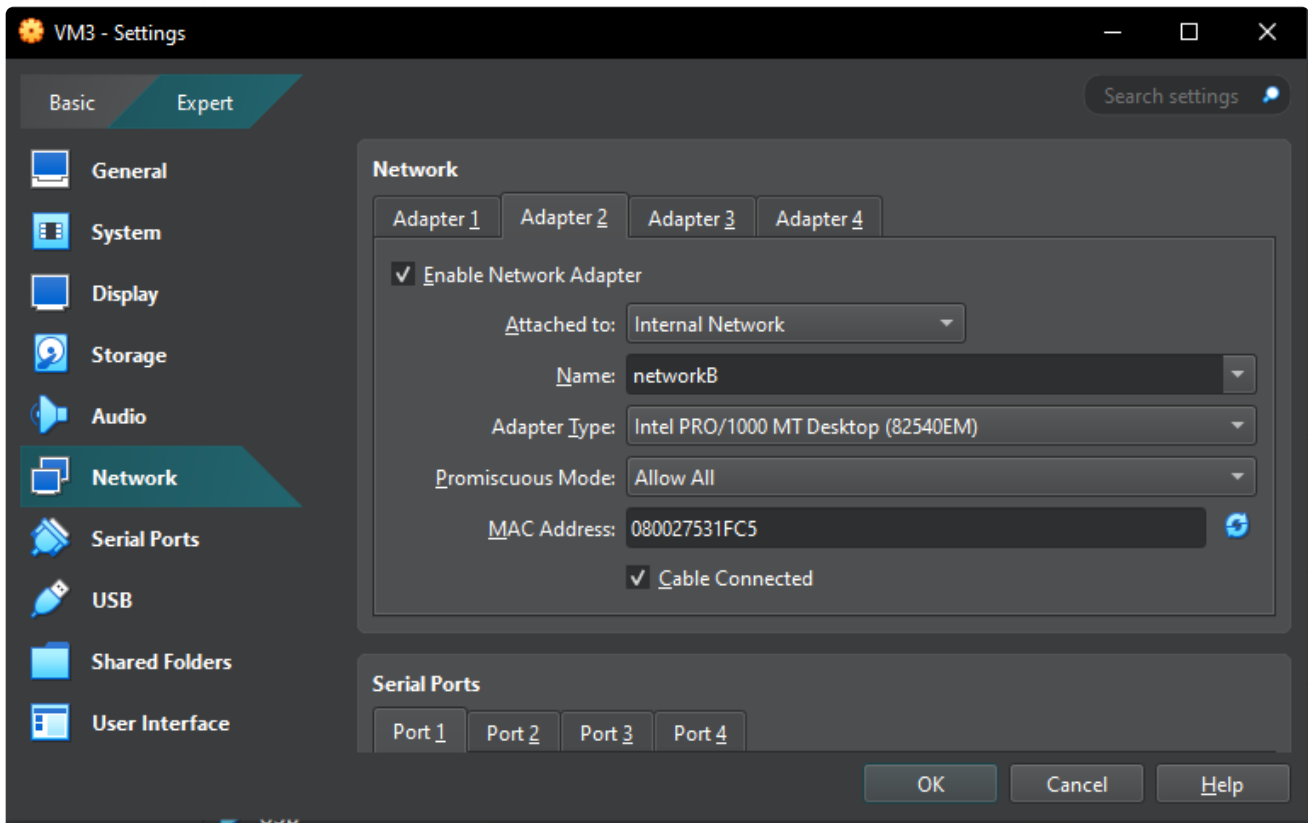
vansil@firewall:~$ ls
vansil@firewall:~$ cat /etc/pf.conf

vansil@firewall:~$ ping 192.168.0.0
PING 192.168.0.0 (192.168.0.0): 56 data bytes
64 bytes from 192.168.0.0: icmp_seq=0 ttl=64 time=3.257 ms
64 bytes from 192.168.0.0: icmp_seq=1 ttl=64 time=3.219 ms
64 bytes from 192.168.0.0: icmp_seq=2 ttl=64 time=3.078 ms
^C
--- 192.168.0.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.078/3.184/3.257/0.077 ms
vansil@firewall:~$ sudo sysrc pf_enable="NO"
pf_enable: YES -> NO
vansil@firewall:~$ sudo sysrc pflog_enable="NO"
pflog_enable: YES -> NO
vansil@firewall:~$ sudo kldload pf
kldload: can't load pf: module already loaded or in kernel
vansil@firewall:~$ sudo pfctl -e
pfctl: pf already enabled
vansil@firewall:~$ sudo /etc/rc.d/pf stop
Cannot 'stop' pf. Set pf_enable to YES in /etc/rc.conf or use 'on
stop' instead of 'stop'.
vansil@firewall:~$ sudo sysrc pf_enable="YES"
pf_enable: NO -> YES
vansil@firewall:~$ sudo sysrc pflog_enable="YES"
pflog_enable: NO -> YES
vansil@firewall:~$ sudo /etc/rc.d/pf stop
Disabling pf.
vansil@firewall:~$

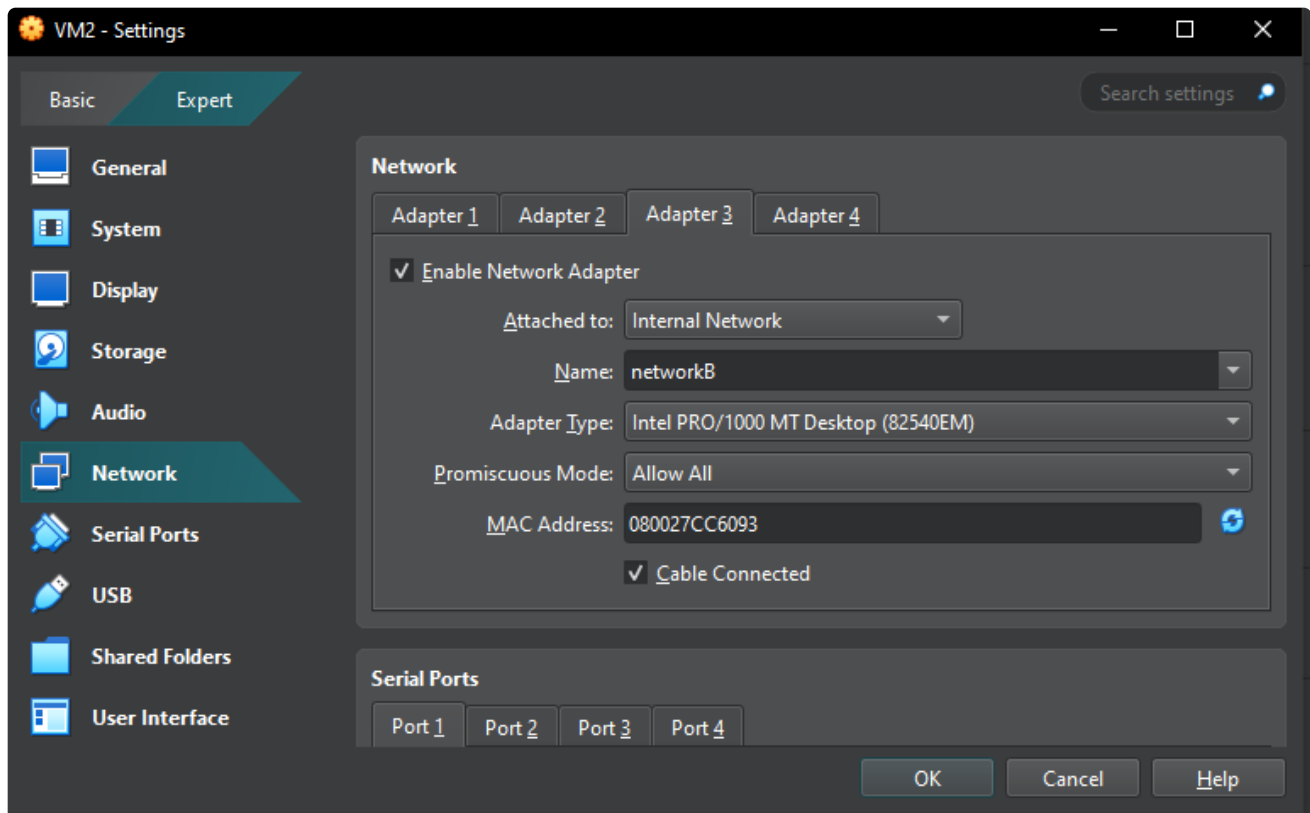
```

Linking VM2 and VM3

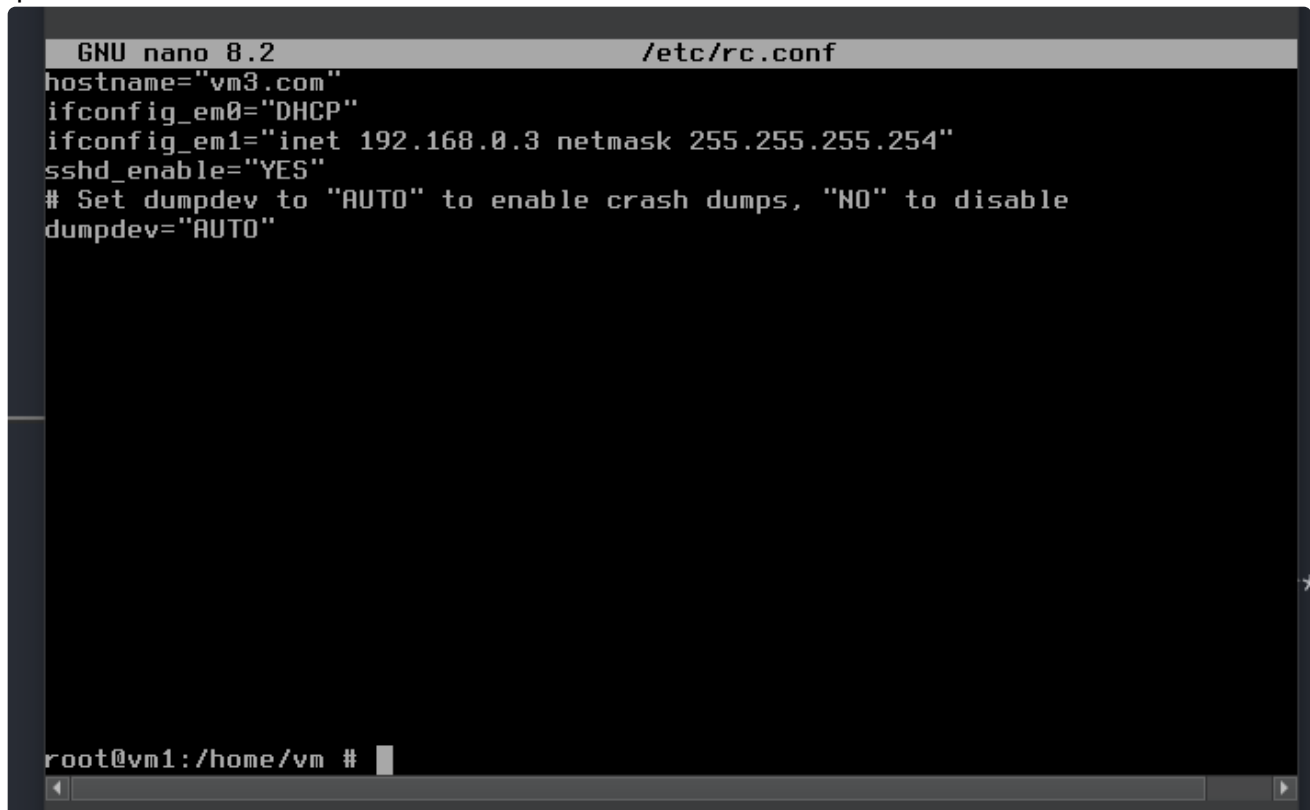
- VM3



- VM2



- Ipaddress in VM3



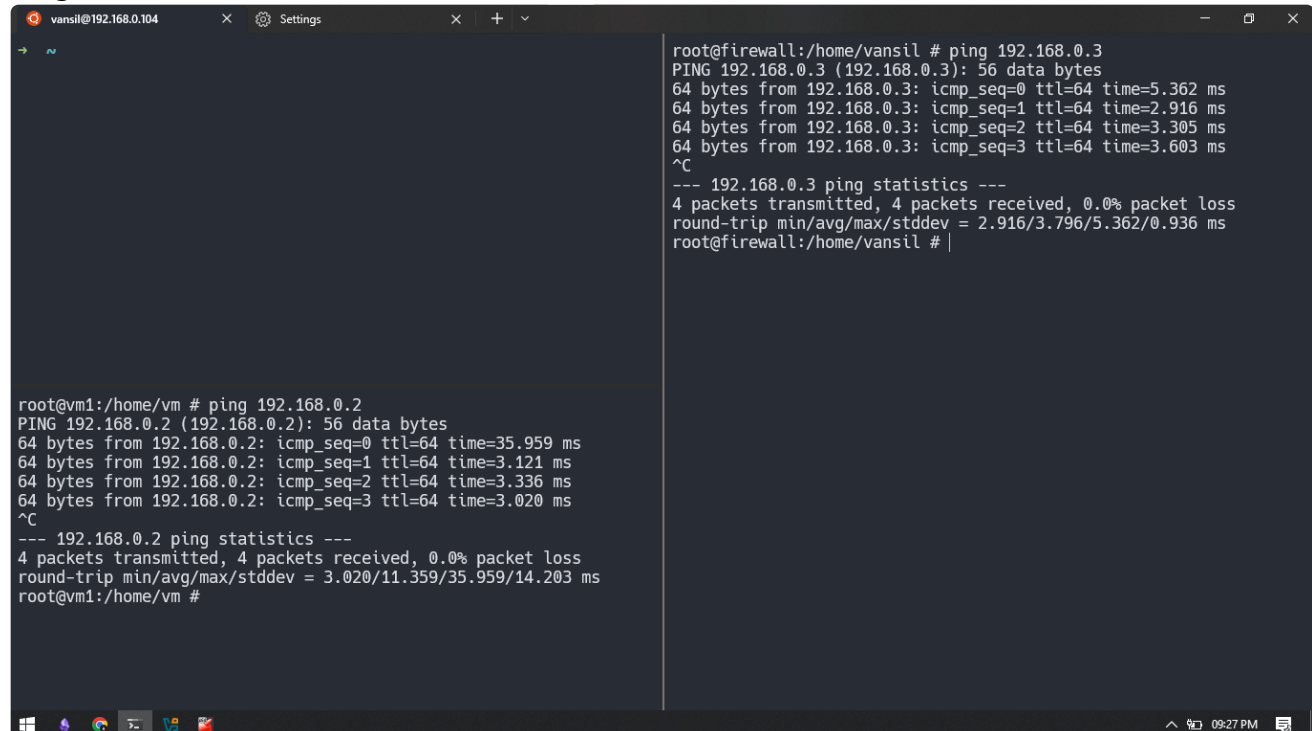
- VM2

```
delete host ::1: gateway lo0
delete net fe80:: gateway ::1
delete net ff02:: gateway ::1
delete net ::ffff:0.0.0.0: gateway ::1
delete net ::0.0.0.0: gateway ::1
add host 127.0.0.1: gateway lo0
add host ::1: gateway lo0
add net fe80:: gateway ::1
add net ff02:: gateway ::1
add net ::ffff:0.0.0.0: gateway ::1
add net ::0.0.0.0: gateway ::1
root@firewall:~ # cat /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.0.1 netmask 255.255.255.254"
ifconfig_em2="inet 192.168.0.2 netmask 255.255.255.254"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

sshd_enable="YES" # enable ssh
pf_enable="YES"   # enable pf (load module if required)
pflog_enable="YES" # start pflogd(8)

root@firewall:~ #
```

- Ping between VM2 and VM3



```
vansil@192.168.0.104 root@vm1:/home/vm # ping 192.168.0.2
PING 192.168.0.2 (192.168.0.2): 56 data bytes
64 bytes from 192.168.0.2: icmp_seq=0 ttl=64 time=35.959 ms
64 bytes from 192.168.0.2: icmp_seq=1 ttl=64 time=3.121 ms
64 bytes from 192.168.0.2: icmp_seq=2 ttl=64 time=3.336 ms
64 bytes from 192.168.0.2: icmp_seq=3 ttl=64 time=3.020 ms
^C
--- 192.168.0.2 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.020/11.359/35.959/14.203 ms
root@vm1:/home/vm #

root@firewall:/home/vansil # ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: icmp_seq=0 ttl=64 time=5.362 ms
64 bytes from 192.168.0.3: icmp_seq=1 ttl=64 time=2.916 ms
64 bytes from 192.168.0.3: icmp_seq=2 ttl=64 time=3.305 ms
64 bytes from 192.168.0.3: icmp_seq=3 ttl=64 time=3.603 ms
^C
--- 192.168.0.3 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.916/3.796/5.362/0.936 ms
root@firewall:/home/vansil #
```

Creating IP routing between VM1 to VM3 via VM2

- Current IP addresses of 3 machines

```

vansil@192.168.0.104
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TS04,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:0f:56:6f
inet 192.168.0.0 netmask 0xffffffff broadcast 255.255.255.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
vm@vm1:~ $
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TS04,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:53:1f:c5
inet 192.168.0.3 netmask 0xffffffff broadcast 255.255.255.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@vm1:/home/vm #
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TS04,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:cc:60:93
inet 192.168.0.2 netmask 0xffffffff broadcast 255.255.255.
255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x4
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
pflog0: flags=100<PROMISC> metric 0 mtu 33160
groups: pflog
root@firewall:/home/vansil #

```

Issue: Can ping from **vm1:eth0** to **vm2:eth0** and **vm2:eth1** and **vm3:eth0** but also able to ping **vm2:eth0** and **vm3:eth0** and not able to ping **vm1:eth0** and **vm2:eth1**

```

vm@192.168.0.107
root@vm1:/home/vm # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: icmp_seq=0 ttl=64 time=3.372 ms
64 bytes from 192.168.0.1: icmp_seq=1 ttl=64 time=3.429 ms
^C
--- 192.168.0.1 ping statistics ---
2 packets transmitted, 2 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.372/3.401/3.429/0.028 ms
root@vm1:/home/vm # ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
ping: sendto: Host is down
ping: sendto: Host is down
ping: sendto: Host is down
^C
--- 192.168.0.3 ping statistics ---
8 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm #
root@vm1:/home/vm # traceroute 192.168.0.1
traceroute to 192.168.0.1 (192.168.0.1), 64 hops max, 40 byte pack
ets
 1 192.168.0.1 (192.168.0.1) 5.868 ms 4.279 ms 3.496 ms
root@vm1:/home/vm # cat /etc/rc.conf
hostname="vm3.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.0.3 netmask 255.255.255.254"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"
root@vm1:/home/vm # traceroute 192.168.0.2
traceroute to 192.168.0.2 (192.168.0.2), 64 hops max, 40 byte pack
ets
 1 192.168.0.2 (192.168.0.2) 7.608 ms 3.227 ms 4.877 ms
root@vm1:/home/vm #
root@firewall:/home/vansil # ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: icmp_seq=0 ttl=64 time=1.099 ms
64 bytes from 192.168.0.1: icmp_seq=1 ttl=64 time=0.911 ms
64 bytes from 192.168.0.1: icmp_seq=2 ttl=64 time=0.891 ms
^C
--- 192.168.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.891/0.967/1.099/0.094 ms
root@firewall:/home/vansil # ping 192.168.0.0
PING 192.168.0.0 (192.168.0.0): 56 data bytes
64 bytes from 192.168.0.0: icmp_seq=0 ttl=64 time=4.047 ms
64 bytes from 192.168.0.0: icmp_seq=1 ttl=64 time=3.228 ms
64 bytes from 192.168.0.0: icmp_seq=2 ttl=64 time=3.136 ms
^C
--- 192.168.0.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.136/3.470/4.047/0.409 ms
root@firewall:/home/vansil # ping 192.168.0.3
PING 192.168.0.3 (192.168.0.3): 56 data bytes
64 bytes from 192.168.0.3: icmp_seq=0 ttl=64 time=7.400 ms
64 bytes from 192.168.0.3: icmp_seq=1 ttl=64 time=2.829 ms
64 bytes from 192.168.0.3: icmp_seq=2 ttl=64 time=3.743 ms
^C
--- 192.168.0.3 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.829/4.657/7.400/1.975 ms
root@firewall:/home/vansil #

```

- Solved (issue was due to hostel wifi having same ip and other adapter connection)

Connecting two machines:

- updated machine ips (192.168.25.x)

```
vm@192.168.0.107
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TSO4,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:0f:56:6f
inet 192.168.25.0 netmask 0xffffffff broadcast 255.255.255
.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@vm1:/home/vm #

options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TSO4,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:53:1f:c5
inet 192.168.25.3 netmask 0xffffffff broadcast 255.255.255
.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x3
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@vm1:/home/vm #

255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
em1: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TSO4,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:1a:d4:a0
inet 192.168.25.1 netmask 0xffffffff broadcast 255.255.255
.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
em2: flags=8863<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 m
tu 1500
options=48505bb<RXCSUM,TXCSUM,VLAN_MTU,VLAN_HWTAGGING,JUMB
O_MTU,VLAN_HWCSUM,TSO4,LRO,VLAN_HWFILTER,VLAN_HWTSO,NOMAP>
ether 08:00:27:cc:60:93
inet 192.168.25.2 netmask 0xffffffff broadcast 255.255.255
.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_
IPV6>
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x4
inet 127.0.0.1 netmask 0xff000000
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
pflog0: flags=100<PROMISC> metric 0 mtu 33160
groups: pflog
root@firewall:/home/vansil #
```

- Change IP addresses to not include 0 and broadcast addresses.

```
vm@192.168.0.107
GNU nano 8.2 /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

sshd_enable="YES" # enable ssh
pf_enable="YES" # enable pf (load module if required)
pflog_enable="YES" # start pflogd(8)

root@firewall:/home/vansil #

GNU nano 8.2 /etc/rc.conf
hostname="vm1.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.1 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

root@vm1:/home/vm #

GNU nano 8.2 /etc/rc.conf
hostname="vm3.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.6 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

root@vm3:/home/vm #
```

Ping within subnets

- VM1-VM2 and VM2-VM3

```

vansil@192.168.0.104
root@firewall:/home/vansil # ping 192.168.25.1
PING 192.168.25.1 (192.168.25.1): 56 data bytes
64 bytes from 192.168.25.1: icmp_seq=0 ttl=64 time=3.329 ms
64 bytes from 192.168.25.1: icmp_seq=1 ttl=64 time=2.817 ms
64 bytes from 192.168.25.1: icmp_seq=2 ttl=64 time=3.768 ms
64 bytes from 192.168.25.1: icmp_seq=3 ttl=64 time=3.558 ms
^C
--- 192.168.25.1 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.817/3.368/3.768/0.354 ms
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil # ping 192.168.25.6
PING 192.168.25.6 (192.168.25.6): 56 data bytes
64 bytes from 192.168.25.6: icmp_seq=0 ttl=64 time=5.336 ms
64 bytes from 192.168.25.6: icmp_seq=1 ttl=64 time=4.370 ms
64 bytes from 192.168.25.6: icmp_seq=2 ttl=64 time=3.317 ms
64 bytes from 192.168.25.6: icmp_seq=3 ttl=64 time=3.988 ms
^C
--- 192.168.25.6 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.317/4.253/5.336/0.730 ms
root@firewall:/home/vansil #

root@vm1:/home/vm # ping 192.168.25.2
PING 192.168.25.2 (192.168.25.2): 56 data bytes
64 bytes from 192.168.25.2: icmp_seq=0 ttl=64 time=22.519 ms
64 bytes from 192.168.25.2: icmp_seq=1 ttl=64 time=3.174 ms
64 bytes from 192.168.25.2: icmp_seq=2 ttl=64 time=4.132 ms
64 bytes from 192.168.25.2: icmp_seq=3 ttl=64 time=3.218 ms
^C
--- 192.168.25.2 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.174/8.261/22.519/8.241 ms
root@vm1:/home/vm #

root@vm3:/home/vm # ping 192.168.25.5
PING 192.168.25.5 (192.168.25.5): 56 data bytes
64 bytes from 192.168.25.5: icmp_seq=0 ttl=64 time=35.016 ms
64 bytes from 192.168.25.5: icmp_seq=1 ttl=64 time=3.494 ms
64 bytes from 192.168.25.5: icmp_seq=2 ttl=64 time=3.735 ms
64 bytes from 192.168.25.5: icmp_seq=3 ttl=64 time=2.955 ms
64 bytes from 192.168.25.5: icmp_seq=4 ttl=64 time=3.285 ms
^C
--- 192.168.25.5 ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.955/9.697/35.016/12.662 ms
root@vm3:/home/vm #

```

Create IP routing from **vm1** to **vm3** via **vm2**

Enable IP Forwarding on **vm2**:

- Enable gateway (`/etc/rc.conf`):

```
gateway_enable="YES"
```

- VM2

```

GNU nano 8.2 /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

pf_enable="YES"      # enable pf (load module if required)
pflog_enable="YES"   # start pflogd(8)
gateway_enable="YES"

```

- restart service routing

```
service routing restart
```

- VM2

```
root@firewall:/home/vansil # service routing restart
delete host 127.0.0.1: gateway lo0
delete host ::1: gateway lo0
delete net fe80::: gateway ::1
delete net ff02::: gateway ::1
delete net ::ffff:0.0.0.0: gateway ::1
delete net ::0.0.0.0: gateway ::1
add host 127.0.0.1: gateway lo0
Additional inet routing options: gateway=YES.
add host ::1: gateway lo0
add net fe80::: gateway ::1
add net ff02::: gateway ::1
add net ::ffff:0.0.0.0: gateway ::1
add net ::0.0.0.0: gateway ::1
root@firewall:/home/vansil # |
```

Set routes on VM1 and VM3

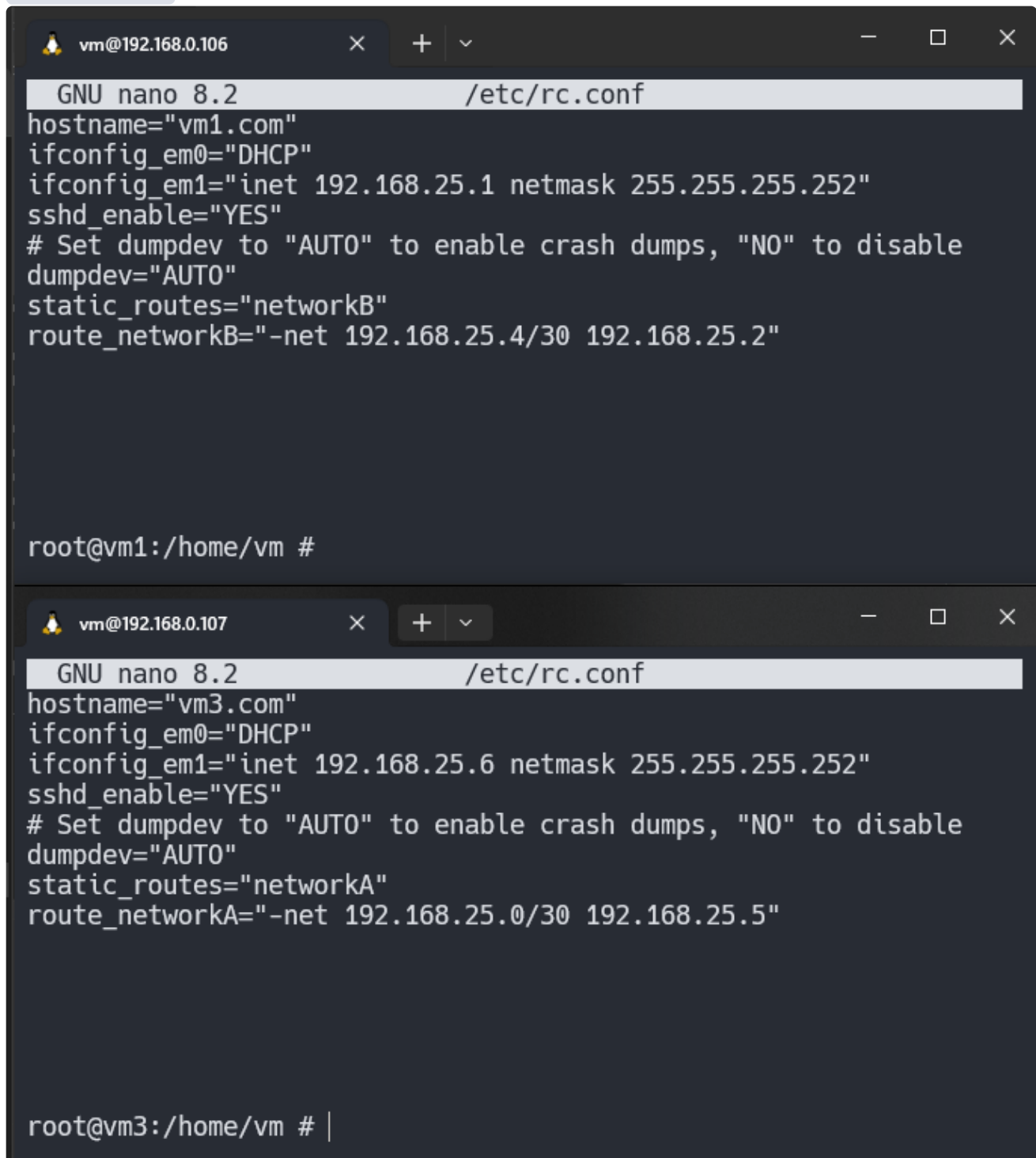
- VM1 (Route to VM3 via VM2)

```
static_routes="networkB"
route_networkB="-net 192.168.25.4/30 192.168.25.2"
```

- VM3 (Route to VM1 via VM2)

```
static_routes="networkA"
route_networkA="-net 192.168.25.0/30 192.168.25.5"
```

- /etc/rc.conf



The image shows two terminal windows side-by-side, both running GNU nano 8.2 to edit /etc/rc.conf. The top window is for VM1 (ip 192.168.0.106) and the bottom window is for VM3 (ip 192.168.0.107). Both configurations set the hostname, network interface (em0), and enable SSH. They also configure static routes for a specific network (192.168.25.4/30 for VM1 and 192.168.25.0/30 for VM3) with a gateway at 192.168.25.2 and 192.168.25.5 respectively. The prompt in the bottom window is root@vm3:/home/vm # |.

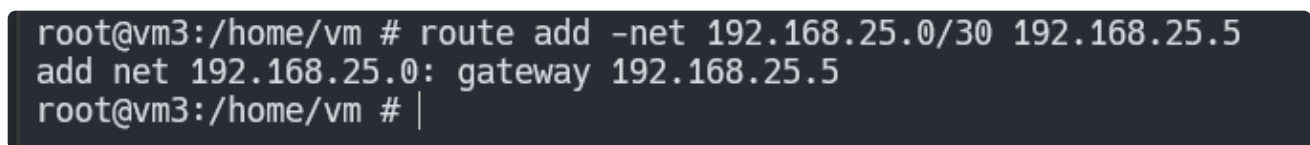
```
vm@192.168.0.106
GNU nano 8.2 /etc/rc.conf
hostname="vm1.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.1 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"
static_routes="networkB"
route_networkB="-net 192.168.25.4/30 192.168.25.2"

root@vm1:/home/vm #

vm@192.168.0.107
GNU nano 8.2 /etc/rc.conf
hostname="vm3.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.6 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"
static_routes="networkA"
route_networkA="-net 192.168.25.0/30 192.168.25.5"

root@vm3:/home/vm # |
```

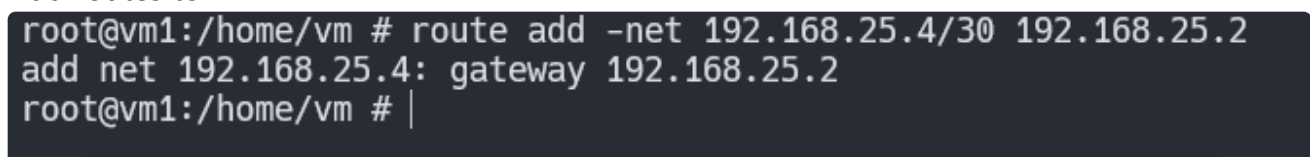
- Add routes to VM3



A terminal window for VM3 showing the command to add a static route for the 192.168.25.0/30 network with a gateway of 192.168.25.5. The prompt is root@vm3:/home/vm # |.

```
root@vm3:/home/vm # route add -net 192.168.25.0/30 192.168.25.5
add net 192.168.25.0: gateway 192.168.25.5
root@vm3:/home/vm # |
```

- Add routes to VM1



A terminal window for VM1 showing the command to add a static route for the 192.168.25.4/30 network with a gateway of 192.168.25.2. The prompt is root@vm1:/home/vm # |.

```
root@vm1:/home/vm # route add -net 192.168.25.4/30 192.168.25.2
add net 192.168.25.4: gateway 192.168.25.2
root@vm1:/home/vm # |
```

- Check routing on VM2

```
netstat -rn
```

- netstat

```
root@firewall:/home/vansil # netstat -rn | grep 192.168.25
192.168.25.0/30      link#2          U              em1
192.168.25.2       link#2          UHS           lo0
192.168.25.4/30    link#3          U              em2
192.168.25.5       link#3          UHS           lo0
root@firewall:/home/vansil # |
```

- ping from VM1 to VM3 and VM3 to VM1

```
vm@192.168.0.106  x  +  v  -  □  ×

--- 192.168.25.6 ping statistics ---
3 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # route add -net 192.168.25.4/30 192.168.25.2
add net 192.168.25.4: gateway 192.168.25.2
root@vm1:/home/vm # ping 192.168.25.6
PING 192.168.25.6 (192.168.25.6): 56 data bytes
64 bytes from 192.168.25.6: icmp_seq=0 ttl=63 time=13.788 ms
64 bytes from 192.168.25.6: icmp_seq=1 ttl=63 time=5.976 ms
64 bytes from 192.168.25.6: icmp_seq=2 ttl=63 time=6.633 ms
64 bytes from 192.168.25.6: icmp_seq=3 ttl=63 time=6.723 ms
^C
--- 192.168.25.6 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.976/8.280/13.788/3.193 ms
root@vm1:/home/vm # |

vm@192.168.0.107  x  +  v  -  □  ×

To quickly create an empty file, use "touch filename".
-- Dru <genesis@istar.ca>
vm@vm3:~ $ su
Password:
root@vm3:/home/vm # route add -net 192.168.25.0/30 192.168.25.5
add net 192.168.25.0: gateway 192.168.25.5
root@vm3:/home/vm # ping 192.168.25.1
PING 192.168.25.1 (192.168.25.1): 56 data bytes
64 bytes from 192.168.25.1: icmp_seq=0 ttl=63 time=5.767 ms
64 bytes from 192.168.25.1: icmp_seq=1 ttl=63 time=6.276 ms
64 bytes from 192.168.25.1: icmp_seq=2 ttl=63 time=6.264 ms
^C
--- 192.168.25.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.767/6.102/6.276/0.237 ms
root@vm3:/home/vm #
```

- traceroute from VM1 to VM3

```
root@vm1:/home/vm # traceroute 192.168.25.6
traceroute to 192.168.25.6 (192.168.25.6), 64 hops max, 40 byte packets
 1  192.168.25.2 (192.168.25.2)  3.293 ms  2.981 ms  4.203 ms
 2  192.168.25.6 (192.168.25.6)  6.316 ms  6.354 ms  6.133 ms
root@vm1:/home/vm # |
```

- traceroute from VM3 to VM1

```
root@vm3:/home/vm # traceroute 192.168.25.1
traceroute to 192.168.25.1 (192.168.25.1), 64 hops max, 40 byte packets
 1  192.168.25.5 (192.168.25.5)  3.310 ms  6.957 ms  4.959 ms
 2  192.168.25.1 (192.168.25.1)  7.034 ms  5.960 ms  7.805 ms
root@vm3:/home/vm # |
```

- Working Finally!

Start Apache HTTP server on VM2

Install Apache

```
pkg install apache24
sysrc apache24_enable="YES"
service apache24 start
```

- Not able to use internet.
- Default Gateway issue.
- Add default gateway
- VM2:

```
root@firewall:/home/vansil # route add default 192.168.0.1
add net default: gateway 192.168.0.1
```

- VM1 and VM3:

```
vm@192.168.0.106
192.168.25.1      link#2      UHS      lo0
192.168.25.4/30  192.168.25.2  UGS      em1
root@vm1:/home/vm # pkg update
Updating FreeBSD repository catalogue...
pkg: An error occurred while fetching package
pkg: An error occurred while fetching package
repository FreeBSD has no meta file, using default settings
pkg: An error occurred while fetching package
pkg: An error occurred while fetching package
pkg: An error occurred while fetching package
pkg: An error occurred while fetching package
Unable to update repository FreeBSD
Error updating repositories!
root@vm1:/home/vm # route add default 192.168.0.1
add net default: gateway 192.168.0.1
root@vm1:/home/vm #

vm@192.168.0.107
root@vm3:/home/vm # netstat -rn | grep 192.
192.168.0.0/24      link#1      U         em0
192.168.0.107      link#1      UHS      lo0
192.168.25.0/30    192.168.25.5  UGS      em1
192.168.25.4/30    link#2      U         em1
192.168.25.6      link#2      UHS      lo0
root@vm3:/home/vm # route 192.168.0.1
route: bad keyword: 192.168.0.1
route: usage: route [-j jail] [-46dnqtv] command [[modifiers] args
]
root@vm3:/home/vm # route add 192.168.0.1
route: writing to routing socket: Invalid argument
add host 192.168.0.1 fib 0: Invalid argument
root@vm3:/home/vm # route add default 192.168.0.1
add net default: gateway 192.168.0.1
root@vm3:/home/vm # |
```

- Apache Installed on VM2

```
=====  
Message from apache24-2.4.62:  
  
--  
To run apache www server from startup, add apache24_enable="yes"  
in your /etc/rc.conf. Extra options can be found in startup script  
.  
  
Your hostname must be resolvable using at least 1 mechanism in  
/etc/nsswitch.conf typically DNS or /etc/hosts or apache might  
have issues starting depending on the modules you are using.  
  
- apache24 default build changed from static MPM to modular MPM  
- more modules are now enabled per default in the port  
- icons and error pages moved from WWWDIR to DATADIR  
  
If build with modular MPM and no MPM is activated in  
httpd.conf, then mpm_prefork will be activated as default  
MPM in etc/apache24/modules.d to keep compatibility with  
existing php/perl/python modules!  
  
Please compare the existing httpd.conf with httpd.conf.sample  
and merge missing modules/instructions into httpd.conf!
```

Enable Apache service at startup

```
sysrc apache24_enable="YES"  
service apache24 start
```

```
root@firewall:/home/vansil # sysrc apache24_enable="YES"  
apache24_enable: -> YES  
root@firewall:/home/vansil # service apache24 start  
Performing sanity check on apache24 configuration:  
Syntax OK  
Starting apache24.  
root@firewall:/home/vansil # |
```

Verify Apache is running

```
service apache24 status
```

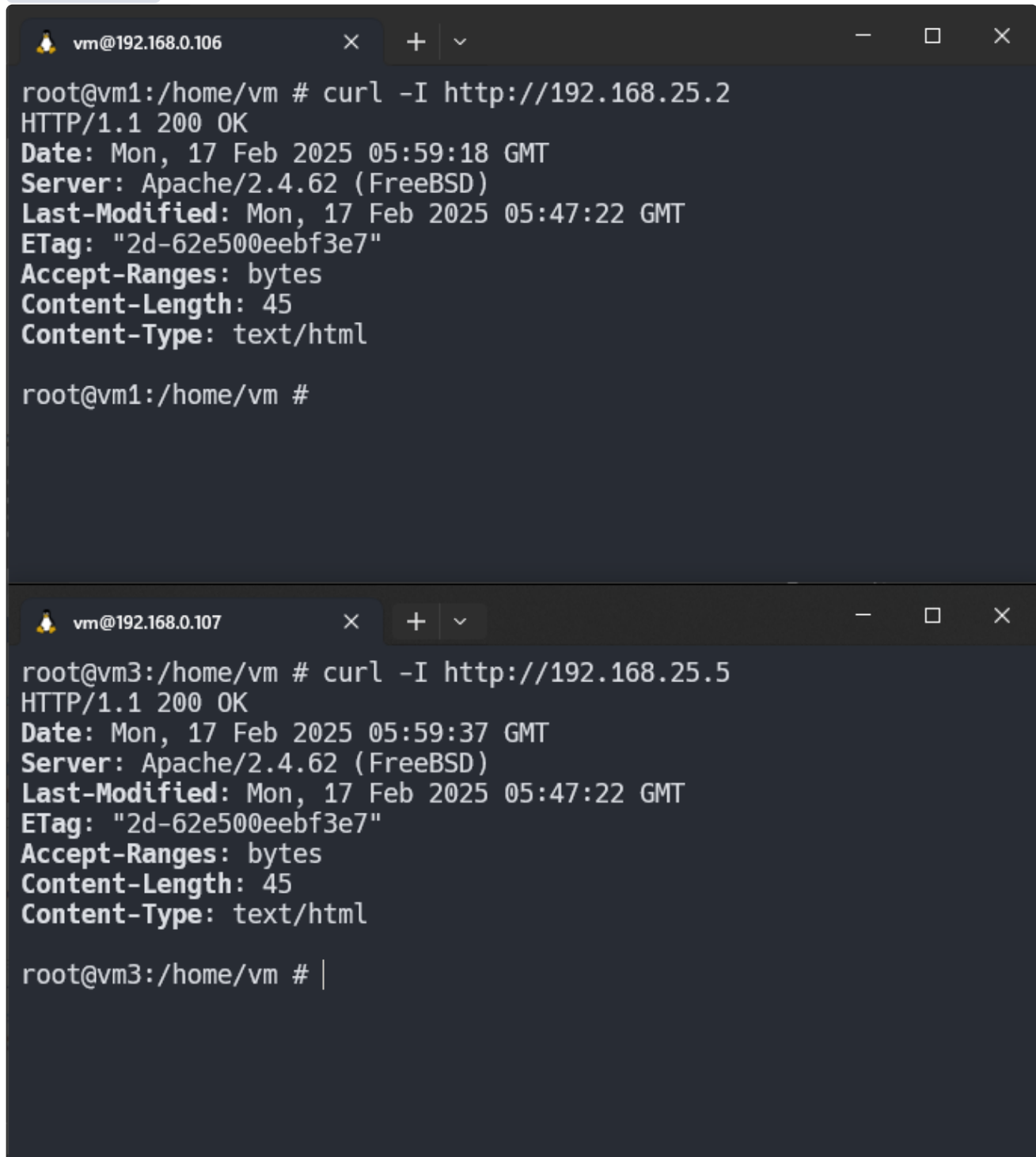
```
root@firewall:/home/vansil # service apache24 status
apache24 is running as pid 3443.
root@firewall:/home/vansil # |
```

Testing using curl from VM1 and VM2

```
curl -I http://192.168.25.2
curl -I http://192.168.25.5
```

- Install curl using `pkg install curl`

- VM1 and VM3:



The image displays two terminal windows side-by-side. The top window, titled 'vm@192.168.0.106', shows a root user at 'vm1:/home/vm' executing a curl command to 'http://192.168.25.2'. The output shows an HTTP 200 OK response with headers including Date, Server (Apache/2.4.62), Last-Modified, ETag, Accept-Ranges, Content-Length (45), and Content-Type (text/html). The bottom window, titled 'vm@192.168.0.107', shows a root user at 'vm3:/home/vm' executing a curl command to 'http://192.168.25.5'. The output is identical, showing an HTTP 200 OK response with the same headers.

```
vm@192.168.0.106
root@vm1:/home/vm # curl -I http://192.168.25.2
HTTP/1.1 200 OK
Date: Mon, 17 Feb 2025 05:59:18 GMT
Server: Apache/2.4.62 (FreeBSD)
Last-Modified: Mon, 17 Feb 2025 05:47:22 GMT
ETag: "2d-62e500eebf3e7"
Accept-Ranges: bytes
Content-Length: 45
Content-Type: text/html

root@vm1:/home/vm #

vm@192.168.0.107
root@vm3:/home/vm # curl -I http://192.168.25.5
HTTP/1.1 200 OK
Date: Mon, 17 Feb 2025 05:59:37 GMT
Server: Apache/2.4.62 (FreeBSD)
Last-Modified: Mon, 17 Feb 2025 05:47:22 GMT
ETag: "2d-62e500eebf3e7"
Accept-Ranges: bytes
Content-Length: 45
Content-Type: text/html

root@vm3:/home/vm # |
```

Create a web directory and add some files.

- Apache's default web root `/usr/local/www/apache24/data`

```
root@firewall:/home/vansil # cd /usr/local/www/apache24
root@firewall:/usr/local/www/apache24 # ls
cgi-bin data      error  icons
root@firewall:/usr/local/www/apache24 # cd data
root@firewall:/usr/local/www/apache24/data # ls
index.html
root@firewall:/usr/local/www/apache24/data # |
```

- create `web` directory:

```
mkdir -p web
```

- VM2:

```
root@firewall:/usr/local/www/apache24/data # mkdir -p web
root@firewall:/usr/local/www/apache24/data # ls
index.html      web
root@firewall:/usr/local/www/apache24/data # tree
.
├── index.html
└── web

2 directories, 1 file
root@firewall:/usr/local/www/apache24/data # |
```

- Add files `index.html`
- Configure Apache to use new directory

```
nano /usr/local/etc/apache24/http.conf
```

- Edit DocumentRoot

```
GNU nano 8.2 /usr/local/etc/apache24/httpd.conf Modified
# documents. By default, all requests are taken from this directory
# symbolic links and aliases may be used to point to other locations
#
DocumentRoot "/usr/local/www/apache24/web"
<Directory "/usr/local/www/apache24/web">
    #
    # Possible values for the Options directive are "None", "All",
    # or any combination of:
    #   Indexes Includes FollowSymLinks SymLinksifOwnerMatch ExecCGI
    #
    # Note that "MultiViews" must be named *explicitly* --- "Options
    # doesn't give it to you.
    #
    # The Options directive is both complicated and important. Please
    # http://httpd.apache.org/docs/2.4/mod/core.html#options
    # for more information.
    #
    Options Indexes FollowSymLinks

    #
    # AllowOverride controls what directives may be placed in .htaccess
    # It can be "All", "None", or any combination of the keywords:
    #   AllowOverride FileInfo AuthConfig Limit
    #
    AllowOverride None

    #
    # Controls who can get stuff from this server.
    #
    Require all granted
</Directory>
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\ Replace	^U Paste	^J Justify

- Restart apache `service apache24 restart`

```
vansil@192.168.0.104 × + v
# Possible values for the Options directive are "None", "All",
# or any combination of:
#   Indexes Includes FollowSymLinks SymLinksifOwnerMatch ExecCGI
#
# Note that "MultiViews" must be named *explicitly* --- "Options MultiViews"
# doesn't give it to you.
#
# The Options directive is both complicated and important. Please consult
# http://httpd.apache.org/docs/2.4/mod/core.html#options
# for more information.
#
Options Indexes FollowSymLinks

#
# AllowOverride controls what directives may be placed in .htaccess files.
# It can be "All", "None", or any combination of the keywords:
#   AllowOverride FileInfo AuthConfig Limit
#
AllowOverride None

#
# Controls who can get stuff from this server.
#
Require all granted
</Directory>

root@firewall:/usr/local/www/apache24/data/web # service apache24
restart
Performing sanity check on apache24 configuration:
AH00526: Syntax error on line 250 of /usr/local/etc/apache24/httpd
.conf:
DocumentRoot '/usr/local/www/apache24/web' is not a directory, or
is not readable
root@firewall:/usr/local/www/apache24/data/web # |
```

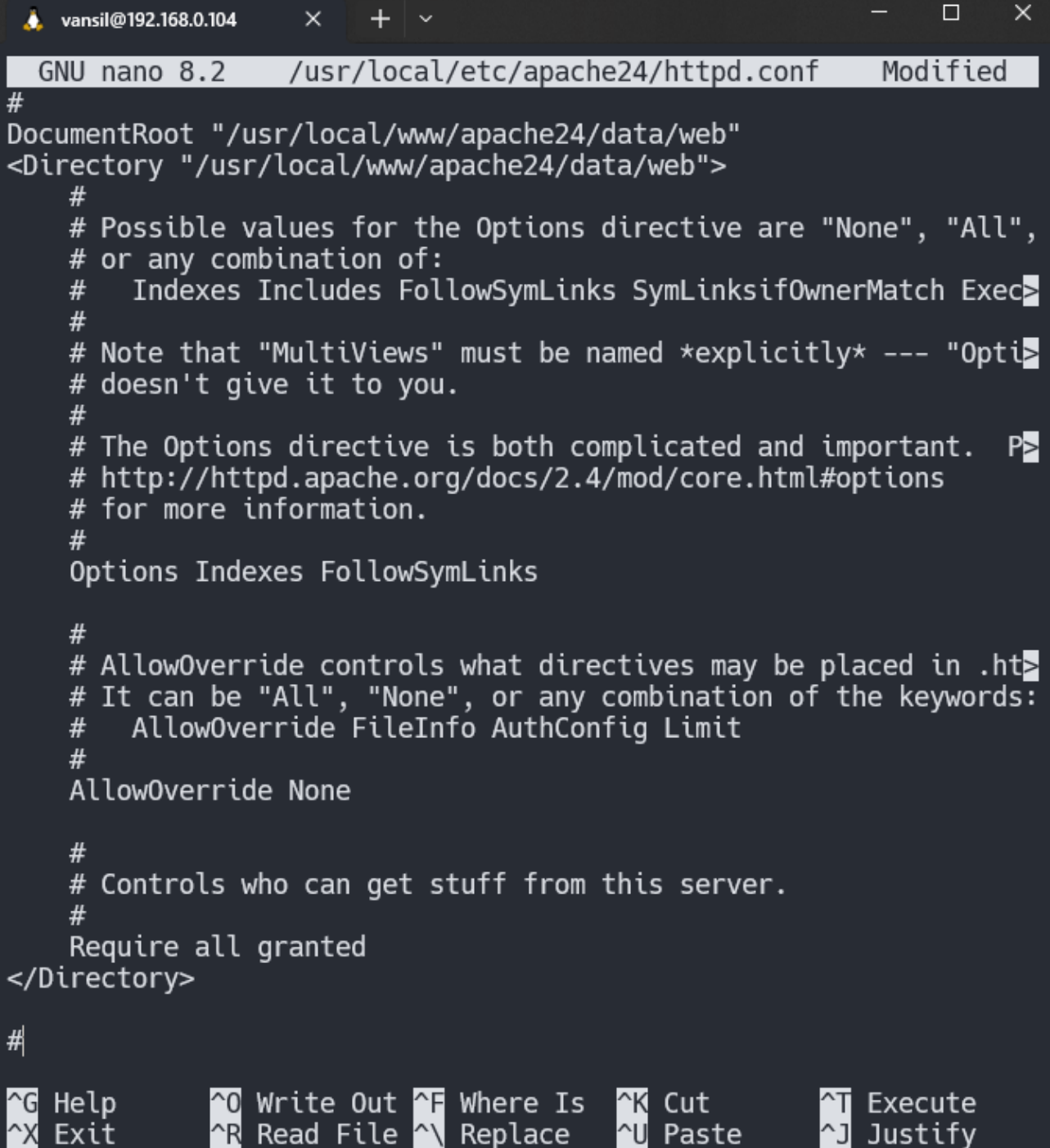
- Directory is not readable.
- Set correct ownership and permissions: [Ownership of Directory](#)

```
chown -R www:www /usr/local/www/apache24/web
chmod -R 755 /usr/local/www/apache24/web
```

- for `web` directory

```
root@firewall:/usr/local/www/apache24/data # chown -R www:www web
root@firewall:/usr/local/www/apache24/data # chmod -R 755 web
root@firewall:/usr/local/www/apache24/data # |
```

- Found error was in path of `DocumentRoot` solved.



```
vansil@192.168.0.104  GNU nano 8.2  /usr/local/etc/apache24/httpd.conf  Modified
#
DocumentRoot "/usr/local/www/apache24/data/web"
<Directory "/usr/local/www/apache24/data/web">
#
# Possible values for the Options directive are "None", "All",
# or any combination of:
#   Indexes Includes FollowSymLinks SymLinksifOwnerMatch Exec
#
# Note that "MultiViews" must be named *explicitly* --- "Opti
# doesn't give it to you.
#
# The Options directive is both complicated and important. P
# http://httpd.apache.org/docs/2.4/mod/core.html#options
# for more information.
#
Options Indexes FollowSymLinks

#
# AllowOverride controls what directives may be placed in .ht
# It can be "All", "None", or any combination of the keywords:
#   AllowOverride FileInfo AuthConfig Limit
#
AllowOverride None

#
# Controls who can get stuff from this server.
#
Require all granted
</Directory>
#

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

- Restart

```
root@firewall:/usr/local/www/apache24/data # service apache24 restart
Performing sanity check on apache24 configuration:
Syntax OK
Stopping apache24.
Waiting for PIDS: 3443.
Performing sanity check on apache24 configuration:
Syntax OK
Starting apache24.
root@firewall:/usr/local/www/apache24/data # |
```

- Check using command `curl http://192.168.25.x/index.html`
- Check in VM1

```
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # |
```

- Check in VM3

```
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # |
```

Run Apache server on port 80:

- Check on which port it is currently running:

```
grep -i "listen" /usr/local/etc/apache24/httpd.conf
```

- Running on port 80 by default

```
root@firewall:/usr/local/www/apache24/data # grep -i "listen" /usr/local/etc/apache24/httpd.conf
# Listen: Allows you to bind Apache to specific IP addresses and/or
# Change this to Listen on specific IP addresses as shown below to
#Listen 12.34.56.78:80
Listen 80
root@firewall:/usr/local/www/apache24/data # |
```


- `/etc/rc.conf`

```
root@firewall:/etc # cat /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

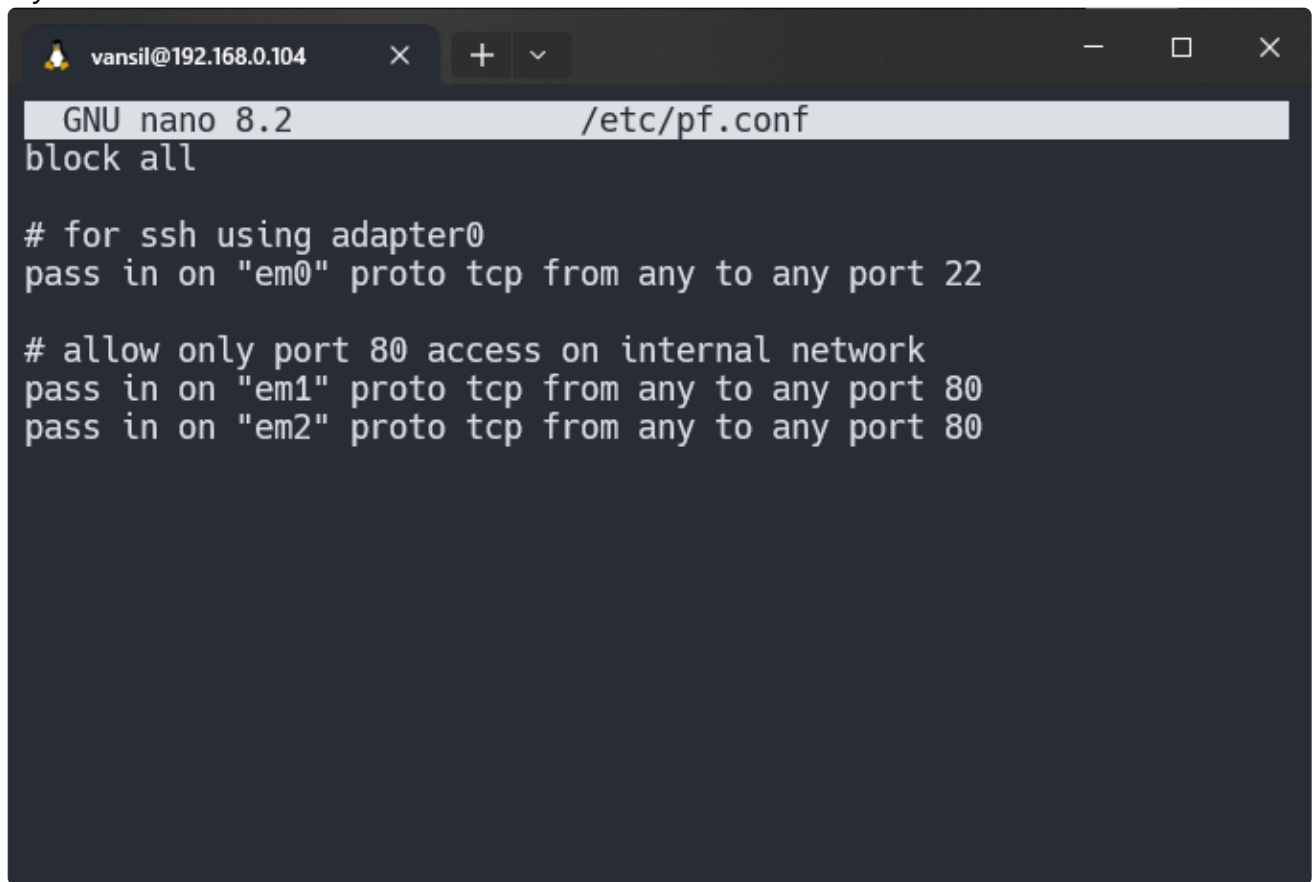
pf_enable="YES"      # enable pf (load module if required)
pflog_enable="YES"   # start pflogd(8)
gateway_enable="YES"
apache24_enable="YES"
root@firewall:/etc # |
```

- It is already enabled.

Configure Rules in `/etc/pf.conf`

- Rules
 - Block all
 - allow ssh 22 only from adapter0
 - from vm1 and vm3 it should be blocked.
 - allow access to port 80

- Try 1

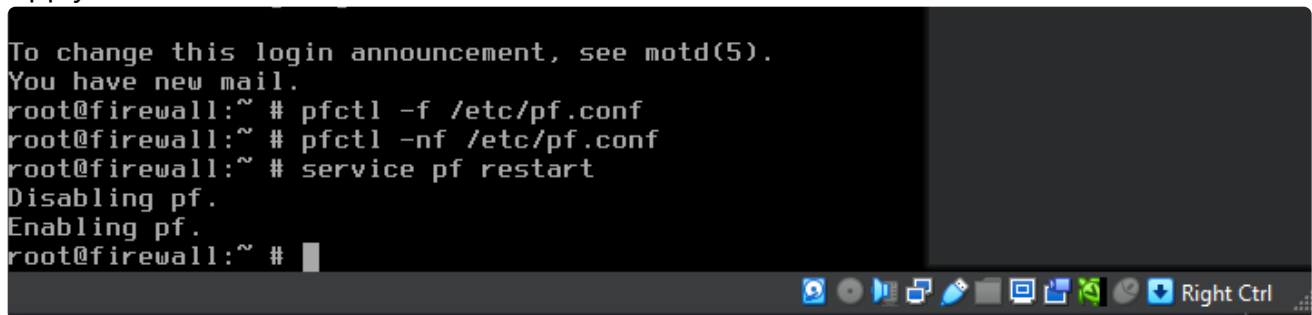


```
vansil@192.168.0.104 x + v
GNU nano 8.2 /etc/pf.conf
block all

# for ssh using adapter0
pass in on "em0" proto tcp from any to any port 22

# allow only port 80 access on internal network
pass in on "em1" proto tcp from any to any port 80
pass in on "em2" proto tcp from any to any port 80
```

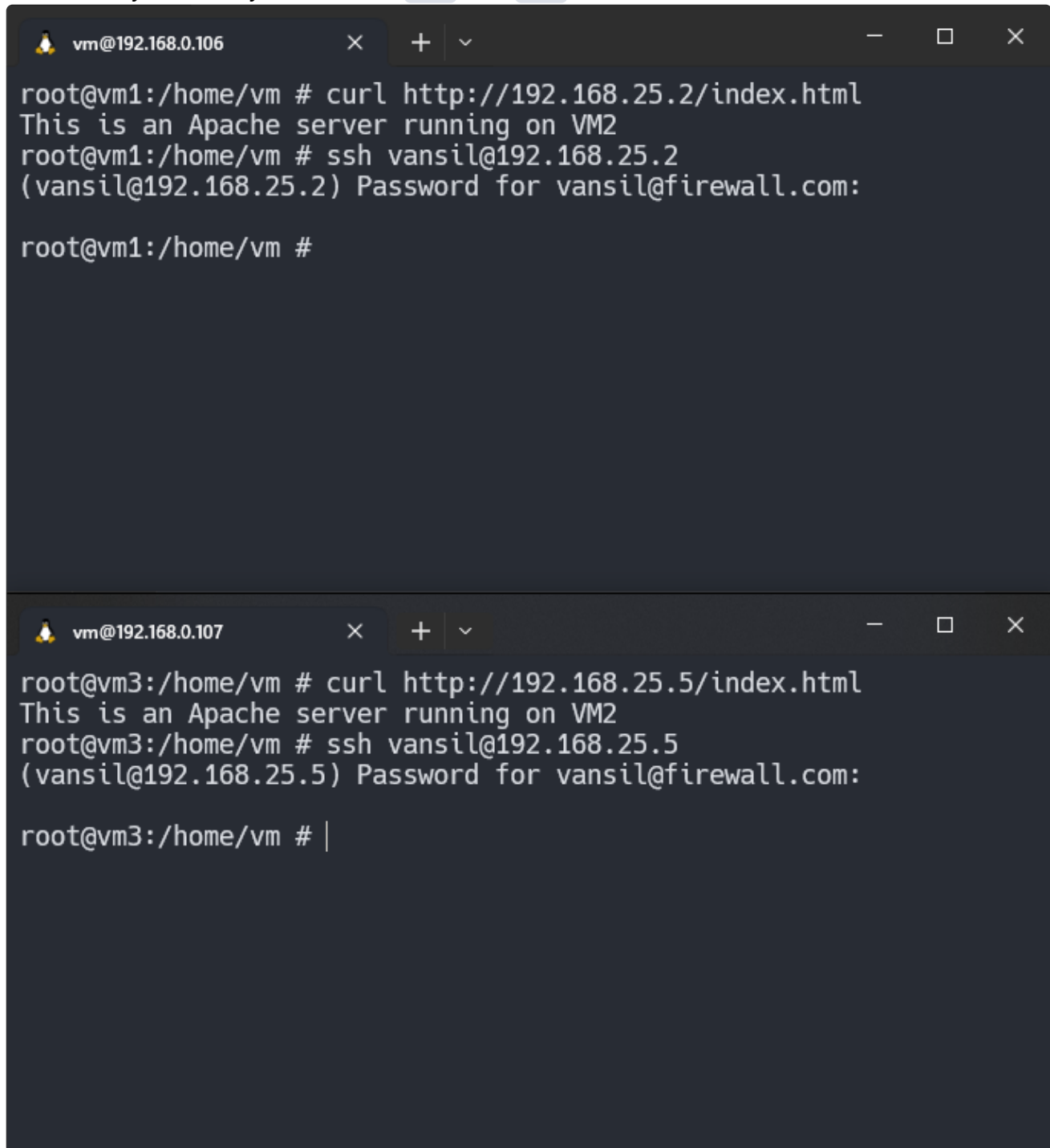
- Apply PF rules



```
To change this login announcement, see motd(5).
You have new mail.
root@firewall:~ # pfctl -f /etc/pf.conf
root@firewall:~ # pfctl -nf /etc/pf.conf
root@firewall:~ # service pf restart
Disabling pf.
Enabling pf.
root@firewall:~ #
```

- Using following rules blocked `icmp` messages as well and not able to detect if error occurred while doing ssh.

- Without any rules... Try to ssh from VM1 and VM3



The image displays two terminal windows stacked vertically. The top window, titled 'vm@192.168.0.106', shows a user at 'root@vm1:/home/vm' running 'curl http://192.168.25.2/index.html', which returns 'This is an Apache server running on VM2'. Subsequently, the user runs 'ssh vansil@192.168.25.2', and the prompt changes to '(vansil@192.168.25.2) Password for vansil@firewall.com:'. The bottom window, titled 'vm@192.168.0.107', shows a user at 'root@vm3:/home/vm' running 'curl http://192.168.25.5/index.html', which also returns 'This is an Apache server running on VM2'. Then, the user runs 'ssh vansil@192.168.25.5', and the prompt changes to '(vansil@192.168.25.5) Password for vansil@firewall.com:'. Both windows have a dark background and a terminal icon in the title bar.

```
vm@192.168.0.106
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # ssh vansil@192.168.25.2
(vansil@192.168.25.2) Password for vansil@firewall.com:

root@vm1:/home/vm #

vm@192.168.0.107
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # ssh vansil@192.168.25.5
(vansil@192.168.25.5) Password for vansil@firewall.com:

root@vm3:/home/vm # |
```

- Before applying firewall... Doing ssh from Main PC, VM1, VM3

```
vansil@192.168.0.104 ~ ssh vansil@192.168.0.104
(vansil@192.168.0.104) Password for vansil@firewall.com:
Last login: Mon Feb 17 12:33:26 2025 from 192.168.0.103
FreeBSD 13.4-RELEASE releng/13.4-n258257-58066db597be GENERIC

Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories: https://www.FreeBSD.org/security/
FreeBSD Handbook: https://www.FreeBSD.org/handbook/
FreeBSD FAQ: https://www.FreeBSD.org/faq/
Questions List: https://www.FreeBSD.org/lists/questions/
FreeBSD Forums: https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with: pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed: freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages: man man
FreeBSD directory layout: man hier

To change this login announcement, see motd(5).
To see how much disk space is left on your UFS partitions, use

    df -h
-- Dru <genesis@istar.ca>
vansil@firewall:~ $
```

```
vm@192.168.0.106 FreeBSD directory layout: man hier

To change this login announcement, see motd(5).
ZFS keeps a history of commands run against a specific pool using the
history subcommand to zpool:

zpool history

More details are available using the -i and -l parameters. Note that
at ZFS will not keep the complete pool history forever and will remove older
events in favor of newer ones.
-- Benedict Reuschling <bcr@FreeBSD.org>
vansil@firewall:~ $
```

```
vm@192.168.0.107 directory, or can be installed later with: pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed: freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages: man man
FreeBSD directory layout: man hier

To change this login announcement, see motd(5).
nc(1) (or netcat) is useful not only for redirecting input/output to
TCP or UDP connections, but also for proxying them with inetd(8).
vansil@firewall:~ $
```

- ssh is working!
- Before applying firewall... Doing curl from Main PC, VM1, VM3

```
~ curl http://192.168.0.104/index.html
This is an Apache server running on VM2
```

```
vm@192.168.0.106 root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm #
```

```
vm@192.168.0.107 root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm #
```

- curl is working for http server.
- Let's apply following rules:

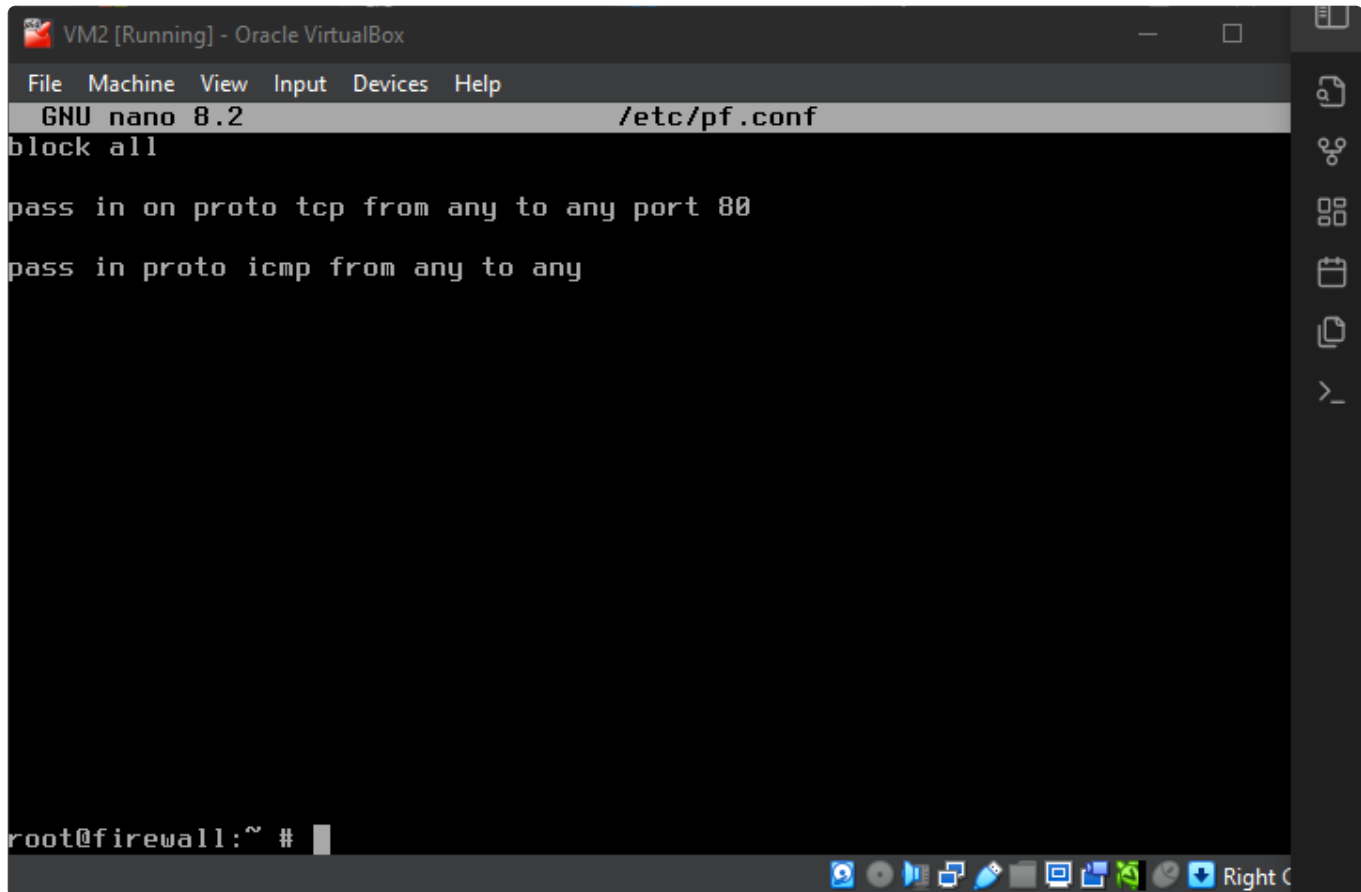
```
block all
```

```
# allow http port 80 for internal network
```

```
pass in on proto tcp from any to any port 80
```

```
# allow internal communication (icmp)
```

```
pass in on proto icmp from any to any
```



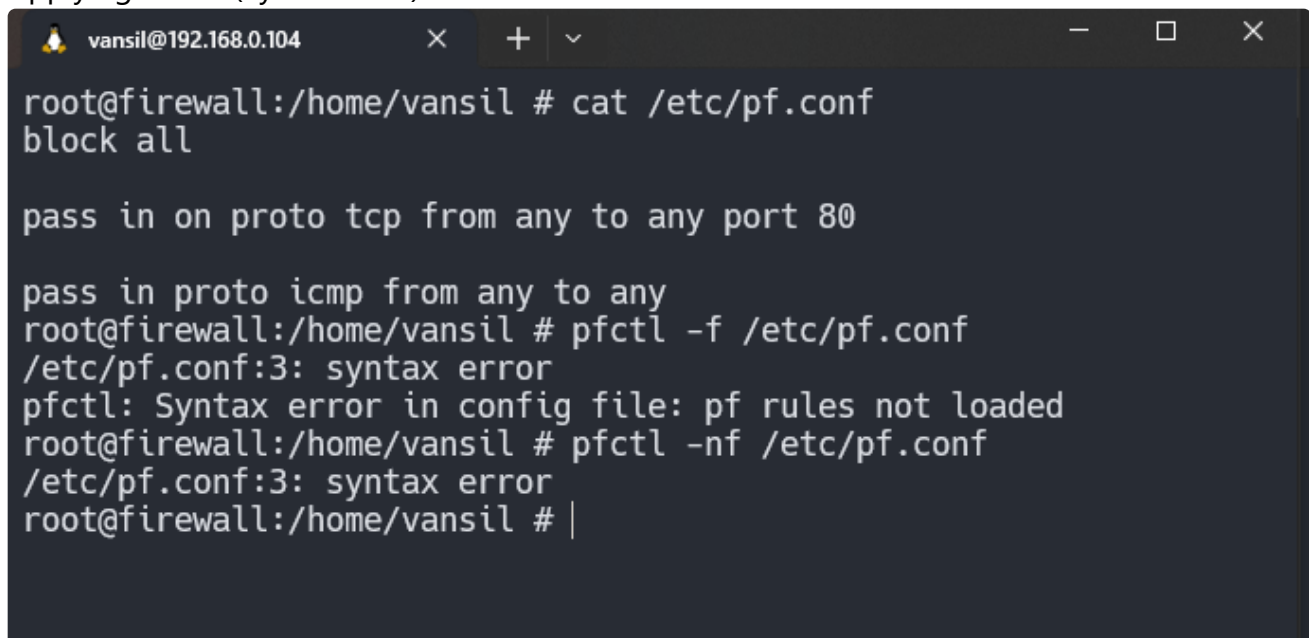
The screenshot shows a VirtualBox window titled "VM2 [Running] - Oracle VirtualBox". Inside, the GNU nano 8.2 text editor is open, editing the file /etc/pf.conf. The content of the file is as follows:

```
block all

pass in on proto tcp from any to any port 80
pass in proto icmp from any to any
```

The terminal prompt at the bottom is `root@firewall:~ #`. The window includes a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". A toolbar on the right side contains icons for file operations. The bottom status bar shows system icons and the text "Right C".

- Applying Rules: (Syntax Error)



The screenshot shows a terminal window with the title bar "vansil@192.168.0.104". The terminal output is as follows:

```
root@firewall:/home/vansil # cat /etc/pf.conf
block all

pass in on proto tcp from any to any port 80

pass in proto icmp from any to any
root@firewall:/home/vansil # pfctl -f /etc/pf.conf
/etc/pf.conf:3: syntax error
pfctl: Syntax error in config file: pf rules not loaded
root@firewall:/home/vansil # pfctl -nf /etc/pf.conf
/etc/pf.conf:3: syntax error
root@firewall:/home/vansil # |
```

- Updated Rules:

```
vansil@192.168.0.104 x + v - □ ×
GNU nano 8.2 /etc/pf.conf
block all

pass in proto tcp from any to any port 80
pass in proto icmp from any to any
```

- The moment Applied PF rules... SSH stuck

```
root@firewall:/home/vansil # pfctl -nf /etc/pf.conf
root@firewall:/home/vansil # pfctl -f /etc/pf.conf
```

- Restart PF service. `service pf restart`
- Try using `curl` to server using Main PC, VM1, VM3
- Try doing ssh from Main PC, VM1, VM3

```
vansil@192.168.0.104 x + v - □ ×
→ ~ curl http://192.168.0.104/index.html
This is an Apache server running on VM2
→ ~ ssh vansil@192.168.0.104
|

vm@192.168.0.106 x + v - □ ×
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # ssh vansil@192.168.25.2

vm@192.168.0.107 x + v - □ ×
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # ssh vansil@192.168.25.5
```

- ssh not working from any machines but `http` server is accessible. Says connection timed out.

```
~ curl http://192.168.0.104/index.html
This is an Apache server running on VM2
~ ssh vasil@192.168.0.104
ssh: connect to host 192.168.0.104 port 22: Connection timed out
~

vm@192.168.0.106
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # ssh vasil@192.168.25.2
ssh: connect to host 192.168.25.2 port 22: Operation timed out
root@vm1:/home/vm #

vm@192.168.0.107
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # ssh vasil@192.168.25.5
ssh: connect to host 192.168.25.5 port 22: Operation timed out
root@vm3:/home/vm #
```

- Do ping work? **yes** because allowed **icmp**

```
~ curl http://192.168.0.104/index.html
This is an Apache server running on VM2
~ ssh vasil@192.168.0.104
ssh: connect to host 192.168.0.104 port 22: Connection timed out
~

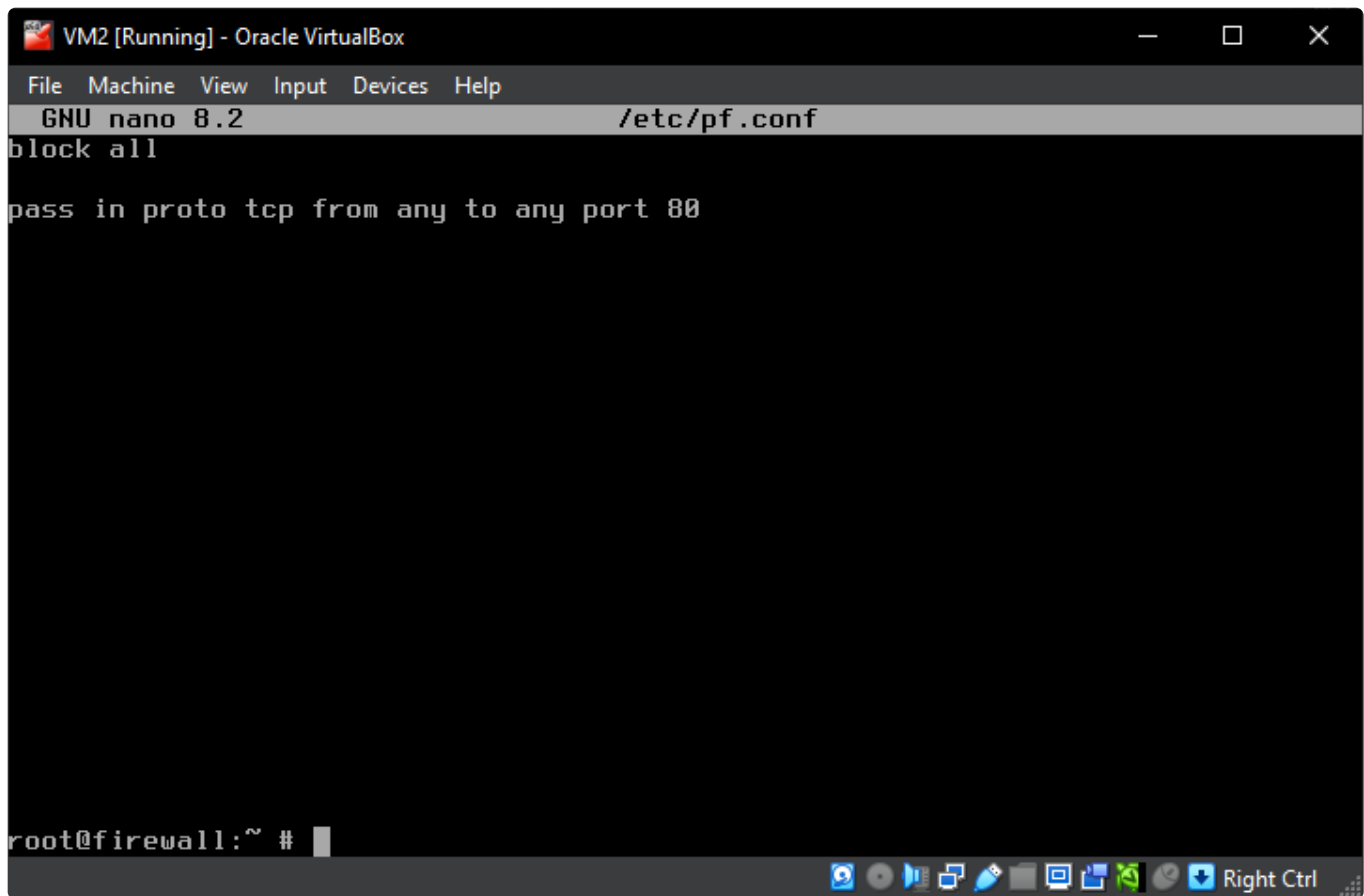
vm@192.168.0.106
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # ssh vasil@192.168.25.2
ssh: connect to host 192.168.25.2 port 22: Operation timed out
root@vm1:/home/vm #

vm@192.168.0.107
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # ssh vasil@192.168.25.5
ssh: connect to host 192.168.25.5 port 22: Operation timed out
root@vm3:/home/vm #
```

- Disable ping.
- New rules:

block all

pass in proto tcp from any to any port 80



```
VM2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 8.2 /etc/pf.conf
block all

pass in proto tcp from any to any port 80

root@firewall:~ #
```

- save and restart pf service.

```
pfctl -nf /etc/pf.conf
pfctl -f /etc/pf.conf
service pf restart
```

```
root@firewall:~ # pfctl -nf /etc/pf.conf
root@firewall:~ # pfctl -f /etc/pf.conf
root@firewall:~ # service pf restart
Disabling pf.
Enabling pf.
root@firewall:~ #
```

- Check ping? Yes ping is also disabled and only **http** server access is allowed.

The image shows two terminal windows. The top window is titled 'vm@192.168.0.106' and shows the following commands and output:

```
root@vm1:/home/vm # ping 192.168.25.2
PING 192.168.25.2 (192.168.25.2): 56 data bytes
^C
--- 192.168.25.2 ping statistics ---
11 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm #
```

The bottom window is titled 'vm@192.168.0.107' and shows the following commands and output:

```
root@vm3:/home/vm # ping 192.168.25.5
PING 192.168.25.5 (192.168.25.5): 56 data bytes
^C
--- 192.168.25.5 ping statistics ---
3 packets transmitted, 0 packets received, 100.0% packet loss
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm #
```

- Final Screenshot: **HTTP** is working anything else is blocked.

The image shows three terminal windows. The top-left window is titled 'vm@192.168.0.106' and shows the following commands and output:

```
root@vm1:/home/vm # ping 192.168.25.2
PING 192.168.25.2 (192.168.25.2): 56 data bytes
^C
--- 192.168.25.2 ping statistics ---
11 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # ssh vansil@192.168.25.2
ssh: connect to host 192.168.25.2 port 22: Operation timed out
root@vm1:/home/vm #
```

The top-right window is titled 'vm@192.168.0.107' and shows the following commands and output:

```
root@vm3:/home/vm # ping 192.168.25.5
PING 192.168.25.5 (192.168.25.5): 56 data bytes
^C
--- 192.168.25.5 ping statistics ---
3 packets transmitted, 0 packets received, 100.0% packet loss
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # ssh vansil@192.168.25.5
ssh: connect to host 192.168.25.5 port 22: Operation timed out
root@vm3:/home/vm #
```

The bottom window is titled 'VM2 [Running] - Oracle VirtualBox' and shows the following commands and output:

```
root@firewall:~ # pfctl -nf /etc/pf.conf
root@firewall:~ # pfctl -f /etc/pf.conf
root@firewall:~ # service pf restart
Disabling pf.
Enabling pf.
root@firewall:~ #
```

---m